EART 22201 Igneous minerals and processes

Igneous processes: Melting



Lesson 5

margaret.hartley@manchester.ac.uk

Igneous processes: Melting

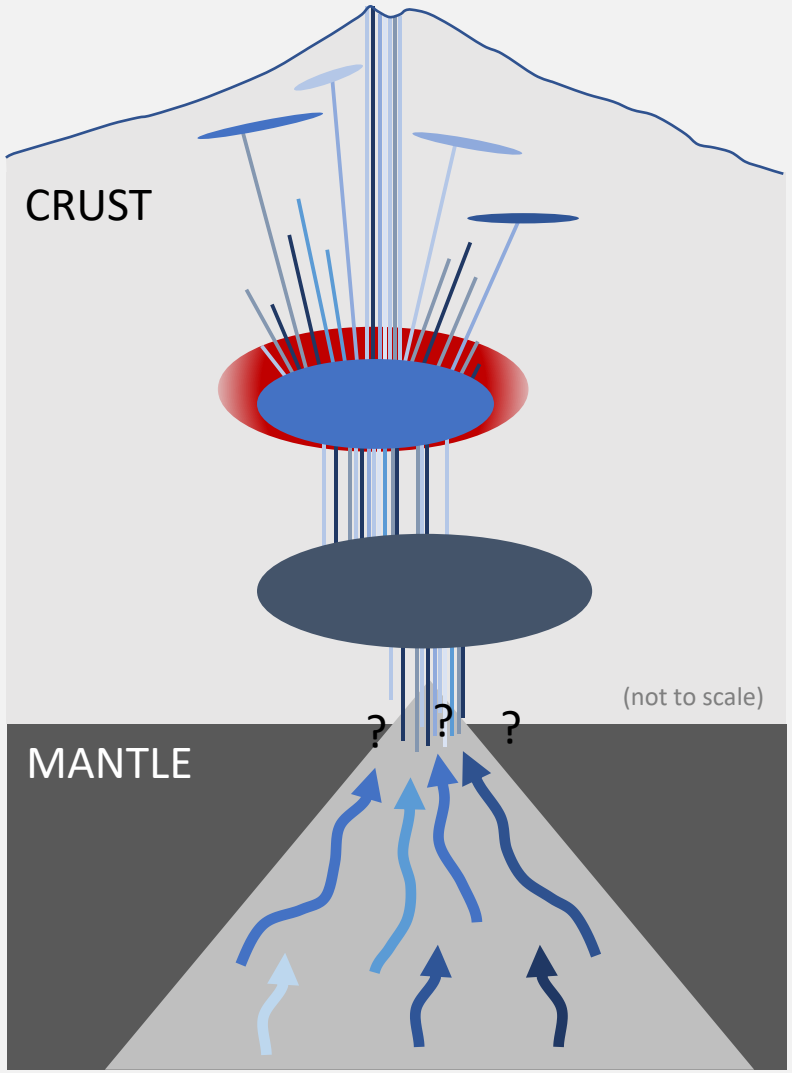
1.1 What melts?

1.2 What is the mantle made of?



margaret.hartley@manchester.ac.uk

What melts? Where does melting happen in the silicate Earth?



Most magmas undergo cooling and crystallization within the Earth's crust, in magma chambers, sills and/or dykes. Some magmas are transported to the Earth's surface where they feed volcanic eruptions.

- Fractional crystallization produces igneous rocks with different chemical compositions and mineral assemblages.
- The rate of crystallization produces different igneous textures.

The ultimate source of most of Earth's magma is **partial melting of the silicate mantle.**

In some cases it is possible to produce magma by partial melting of crustal rocks.

More on crustal melting in weeks 8 and 9

margaret.hartley@manchester.ac.uk

Igneous processes: Melting

1.1 What melts?

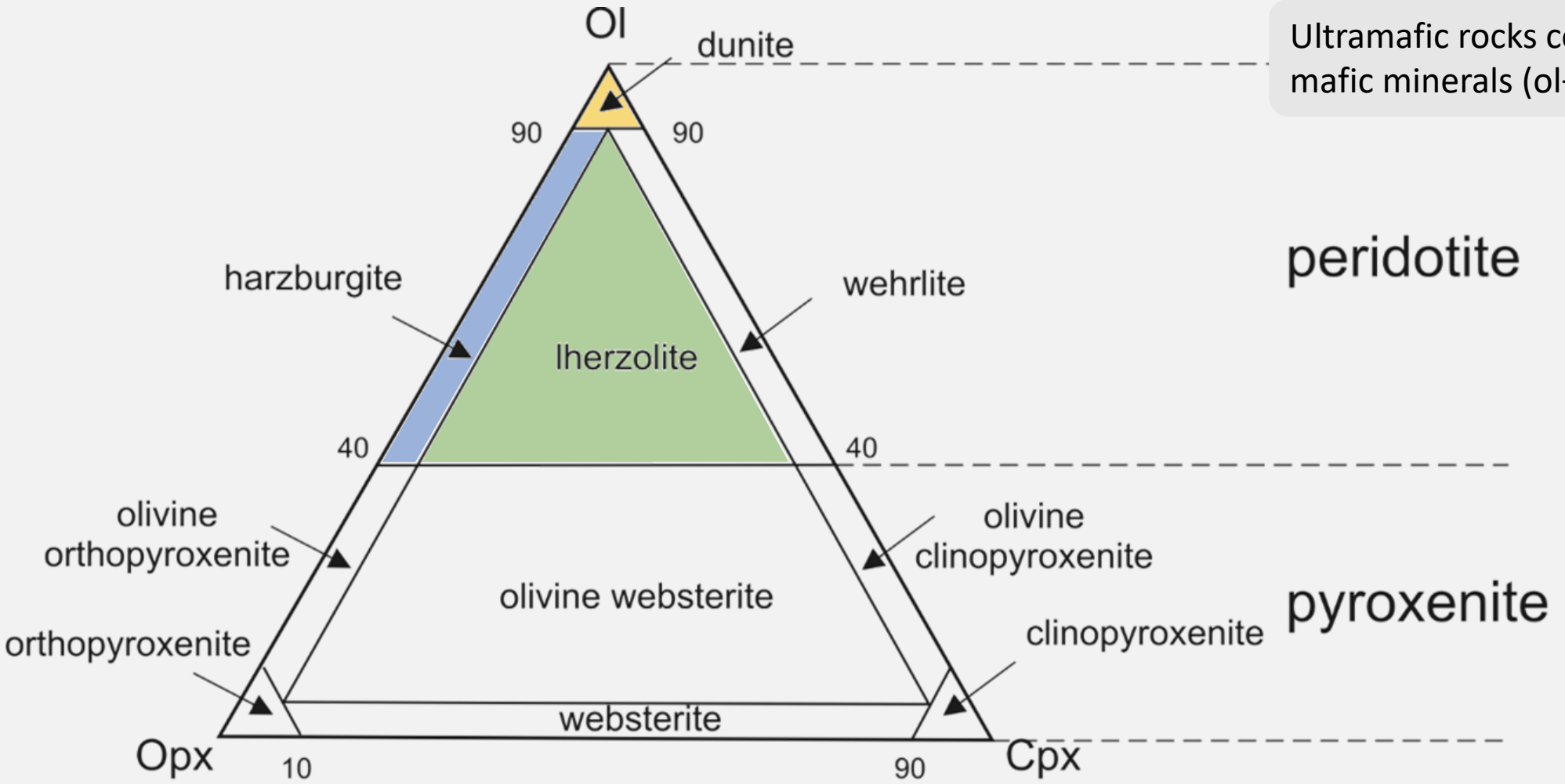
1.2 What is the mantle made of?



margaret.hartley@manchester.ac.uk

What is the mantle made of? Mineralogy and classification of mantle rocks

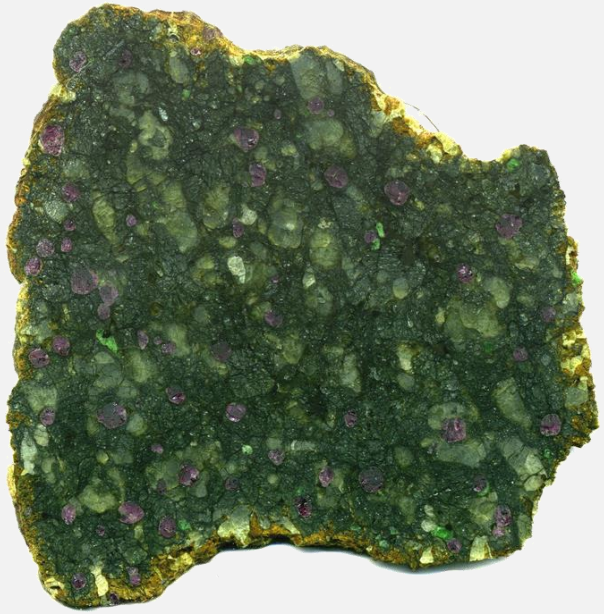
Ultramafic rocks containing >90% mafic minerals (ol+cpx+opx)



margaret.hartley@manchester.ac.uk

Peridotites in hand specimen

Lherzolite



ol + cpx + opx

Harzburgite



ol + opx

Dunite



ol

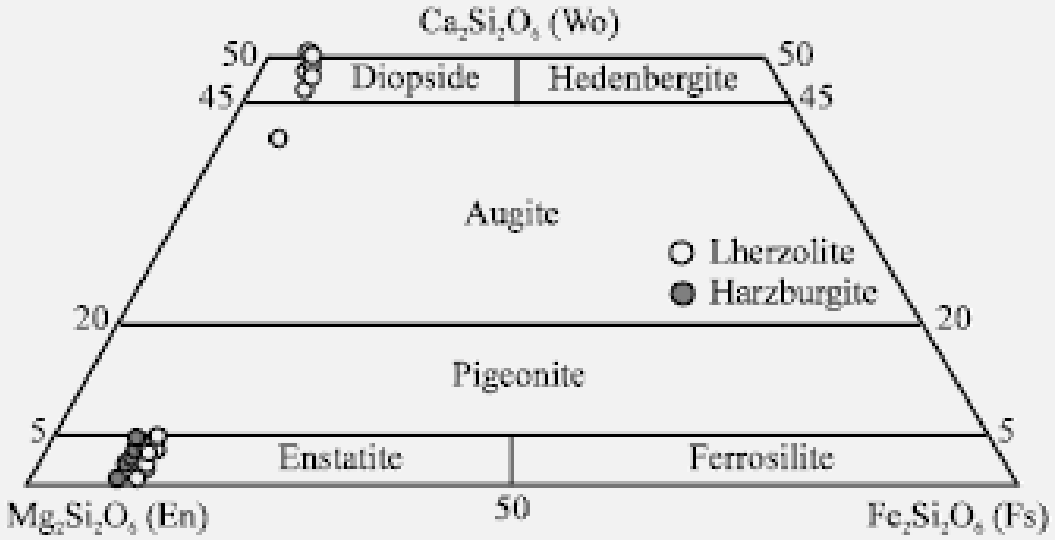
margaret.hartley@manchester.ac.uk

Mineral compositions in mantle peridotite

Mantle minerals are close to the Mg-rich endmembers of their solid solutions.

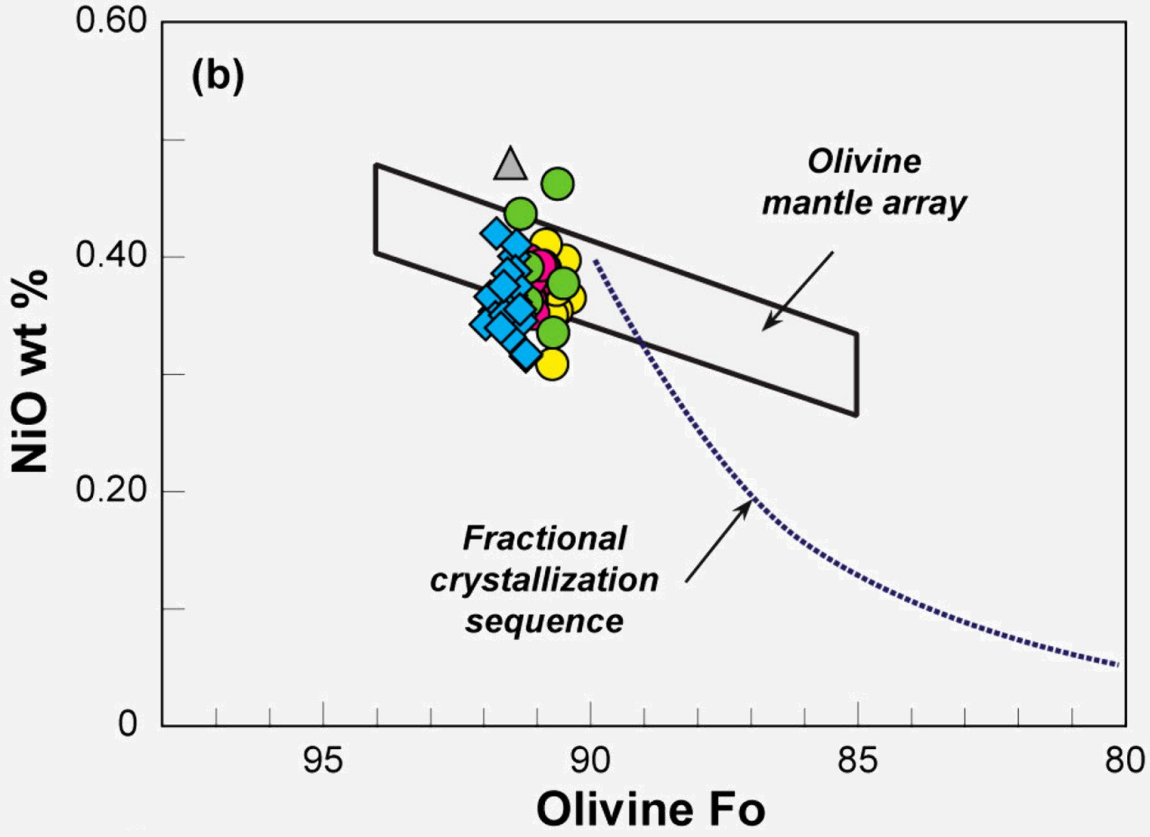
Mg-number = $\left(\frac{Mg^{2+}}{Mg^{2+}+Fe^{2+}}\right)$ ≈ 90-92 in mantle minerals

Example: typical mantle olivine has composition Fo_{92}



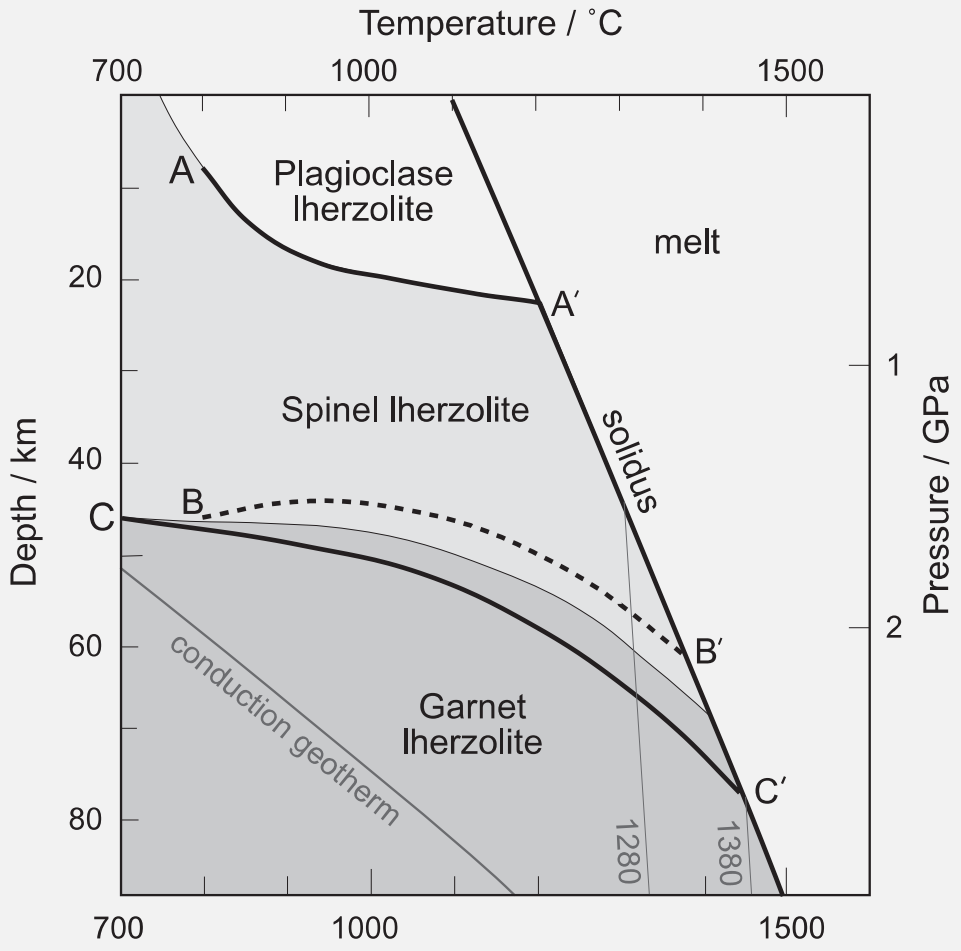
Reminder from Week 3: The pure-Mg endmembers of the olivine and pyroxene solid solutions are:

Olivine	Mg_2SiO_4 (forsterite)
Orthopyroxene	$Mg_2Si_2O_6$ (enstatite)
Clinopyroxene	$CaMgSi_2O_6$ (diopside)



margaret.hartley@manchester.ac.uk

Al-bearing phase in mantle peridotites



Gill (2010) Fig. 5.5

The peridotite mineral assemblage is ol+cpx+opx plus an Al-bearing mineral. The stable Al-bearing mineral depends on pressure:

0 < P < 0.8 GPa

Plagioclase, $CaAl_2Si_2O_8$

plagioclase peridotite is rare in nature because its stability field is usually occupied by the crust

0.8 < P < 2 GPa

Spinel, $MgAl_2O_4$

P > 2 GPa

Garnet, $Mg_3Al_2Si_3O_8$

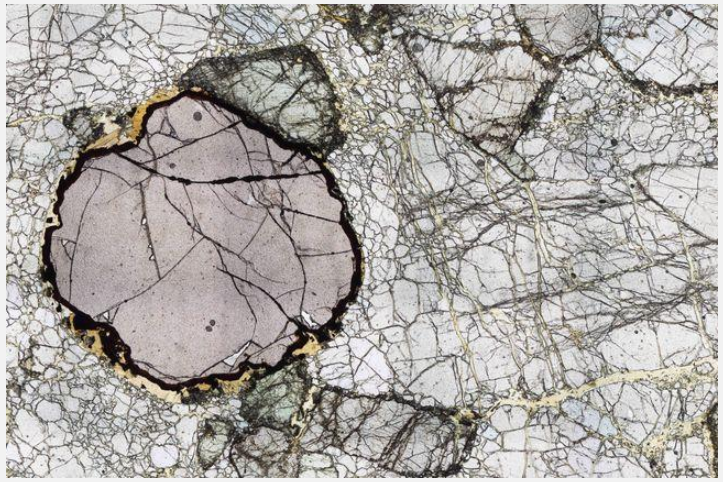
Phlogopite, $KMg_3AlSi_3O_{10}(OH)_2$

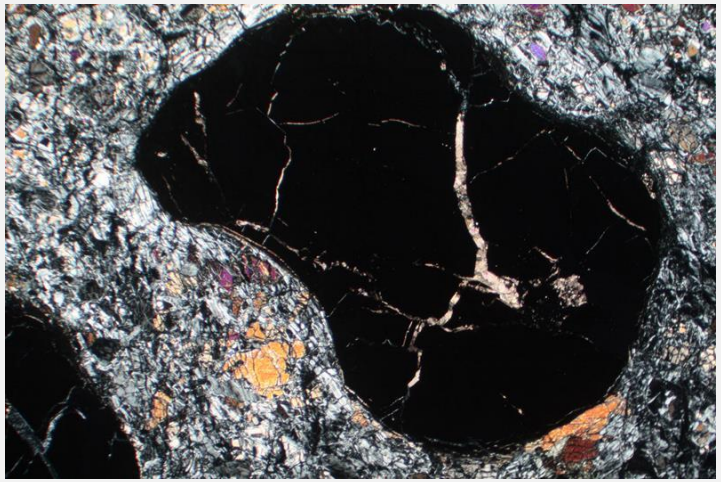
Mg-rich endmember of biotite solid solution. Indicates hydrous mantle

margaret.hartley@manchester.ac.uk

Garnet

See Week 3 Rock-forming minerals





Garnet lherzolite peridotites: pressure > 2 GPa

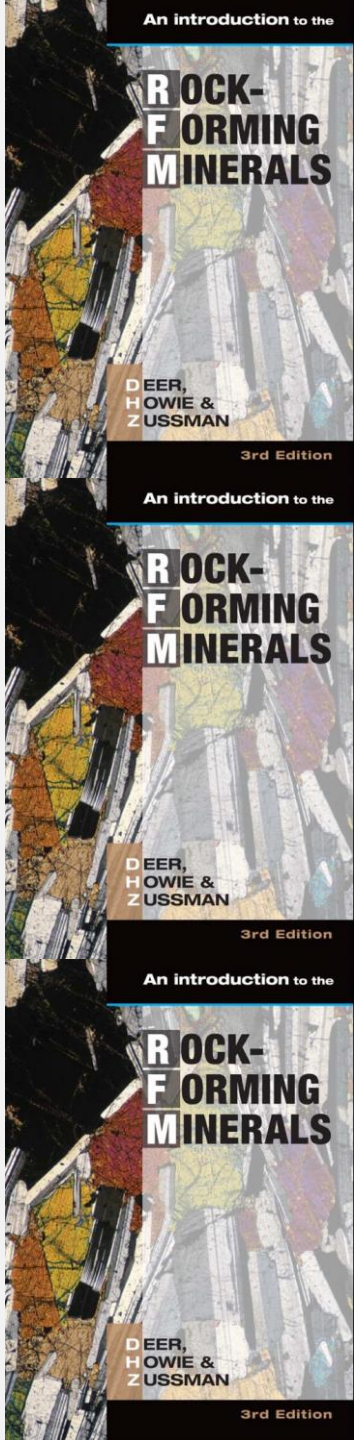
Mantle garnet is typically a solid solution between $Mg_3Al_2Si_3O_{12}$ and $Fe_3Al_2Si_3O_{12}$



Key optical properties:

PPL
- Pinkish colour
- Very high relief
- Fractures (common)
- No cleavage

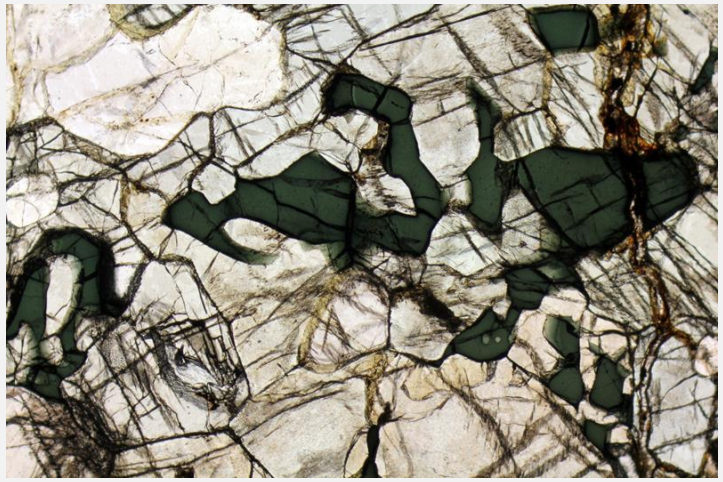
XP
- Isotropic – always in extinction

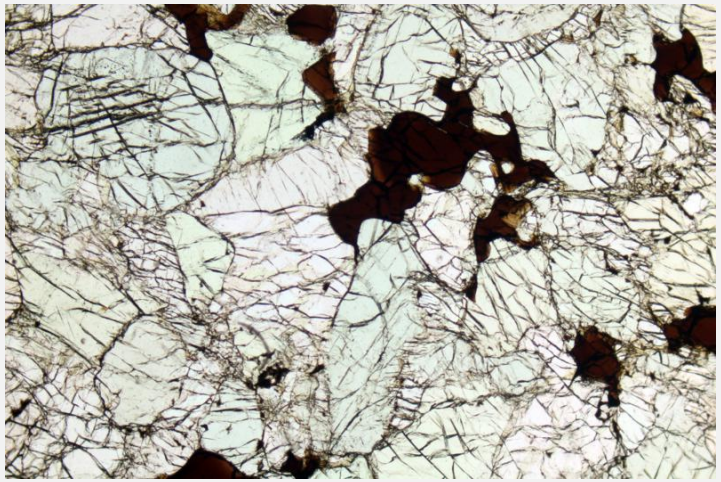


margaret.hartley@manchester.ac.uk

Spinel

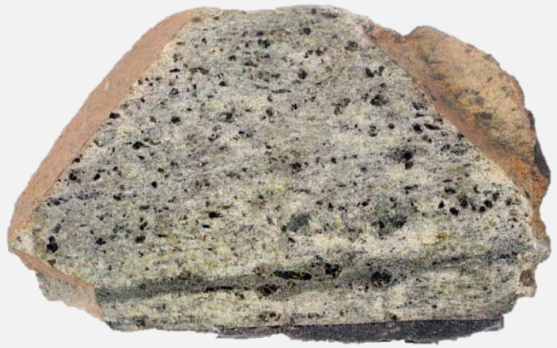
NEW! Rock-forming mineral video in Blackboard Week 5 folder





Spinel lherzolite peridotites: pressure 0.8-2 GPa

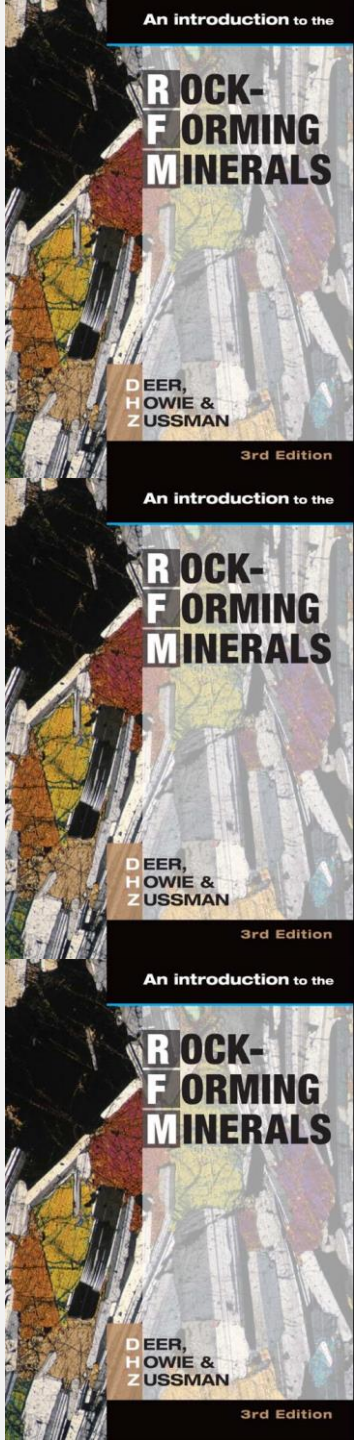
Mantle spinel is typically a solid solution between $MgAl_2O_4$ and $FeAl_2O_4$



Key optical properties:

PPL
- Strongly coloured (often red, brown, green); may appear nearly opaque
- Not pleochroic
- High relief
- No cleavage

XP
- Isotropic – always in extinction
- No zoning



margaret.hartley@manchester.ac.uk

Igneous processes: Melting

You should now be able to:

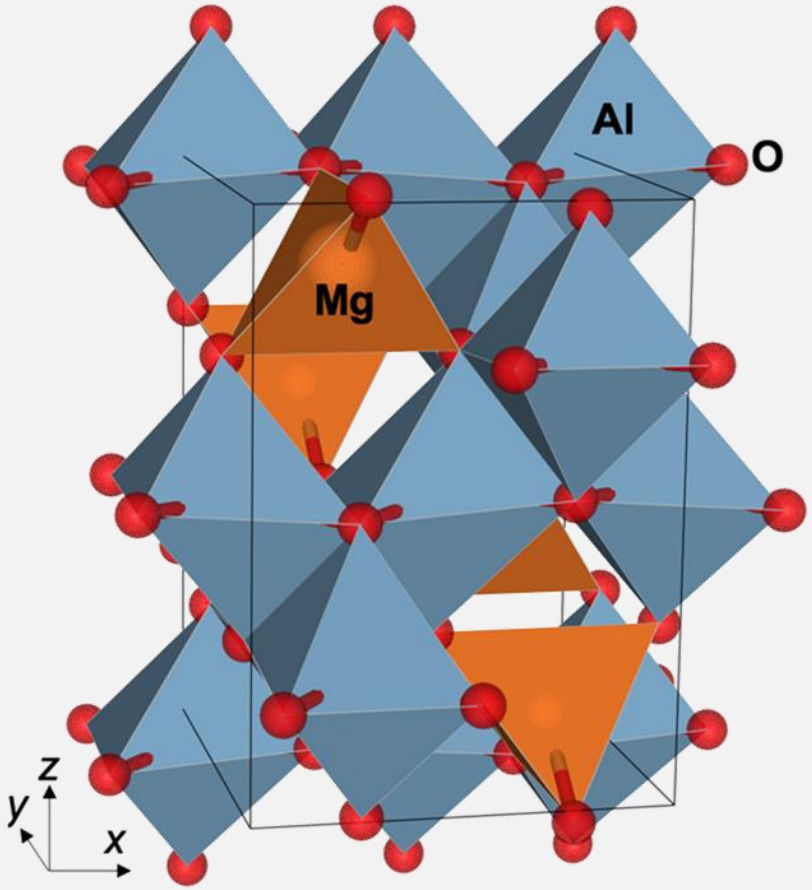
- Describe where melting takes place within the silicate Earth
- State the mineral assemblages of typical mantle rocks, and classify mantle rocks according to their modal mineralogy
- Explain how the mineralogy of mantle rocks varies as a function of pressure
- Recognise the aluminous phase (garnet or spinel) in thin sections of mantle rocks



margaret.hartley@manchester.ac.uk

Spinel group minerals

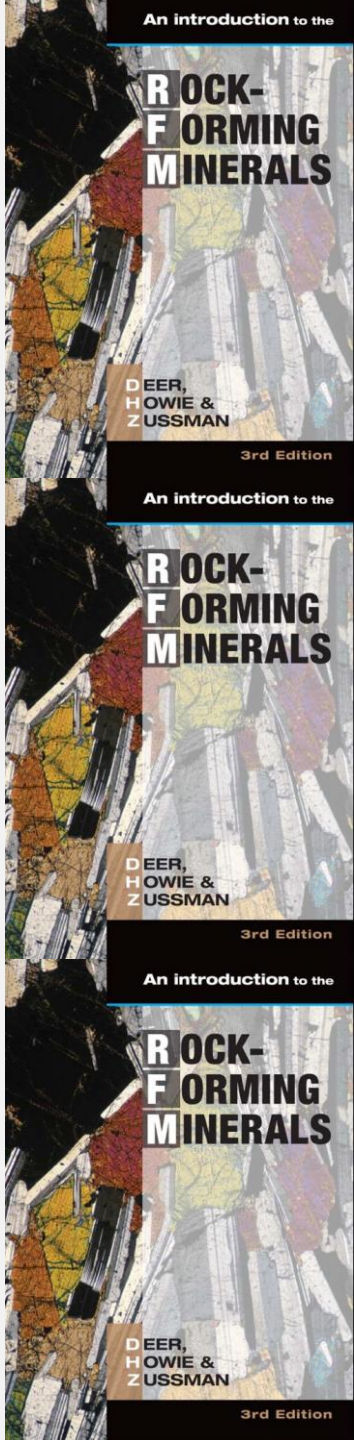
AB_2O_4 oxide (non-silicate) cubic





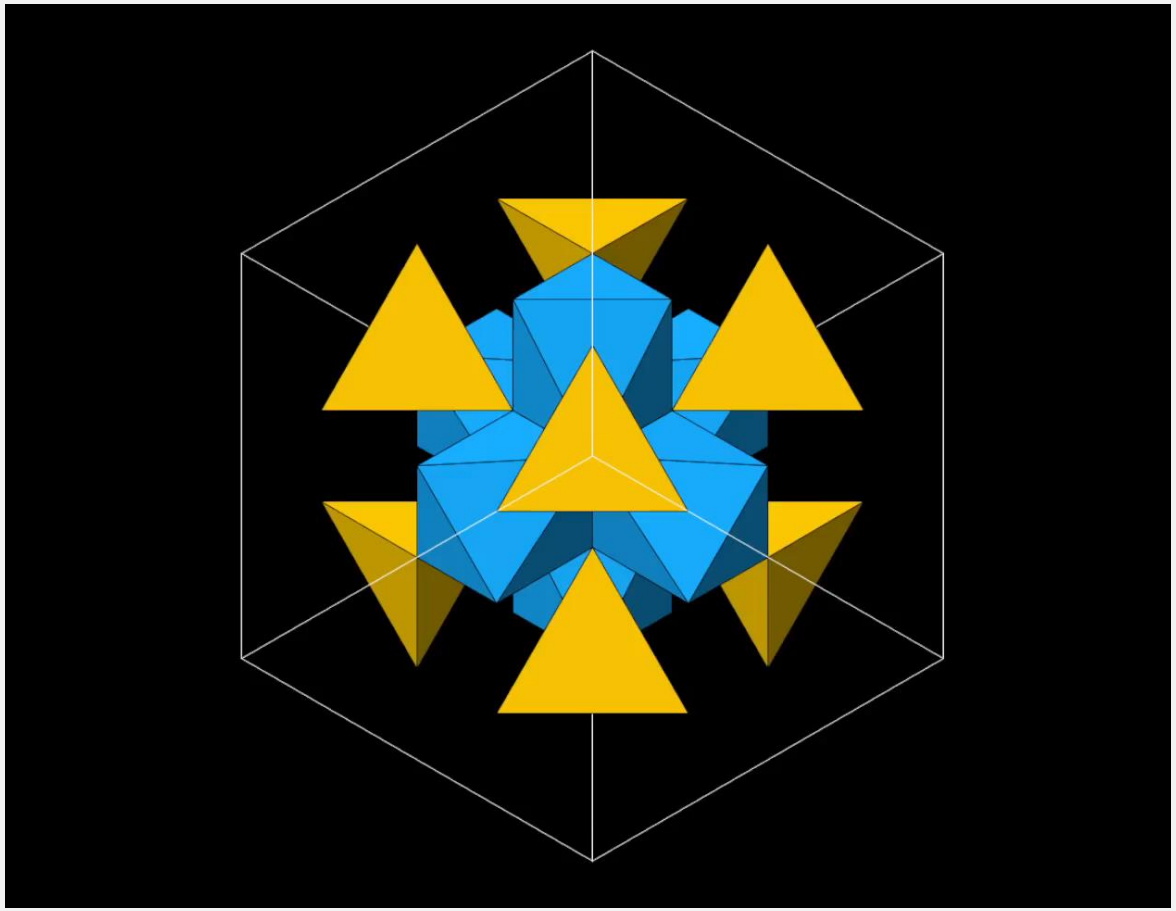


Spinel group minerals are common accessory minerals in a variety of metamorphic and igneous rocks

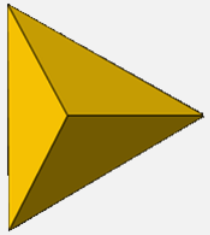


margaret.hartley@manchester.ac.uk

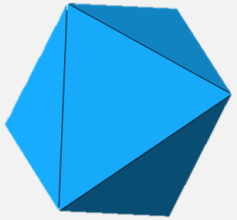
Spinel group minerals



General formula: AB_2O_4



A = Small M^{2+} cation:
Mg^{2+}, Fe^{2+}, Mn^{2+}, Zn^{2+}, Ni^{2+}
Tetrahedral [4] co-ordination



B = Small M^{3+} cation:
Al^{3+}, Fe^{3+}, Cr^{3+}, Ti^{3+}
Octahedral [6] co-ordination

The spinel mineral group includes a wide range of compositions and solid solutions, but all have the same structure.

margaret.hartley@manchester.ac.uk

Spinel *sensu stricto*

$MgAl_2O_4$ oxide (non-silicate) cubic





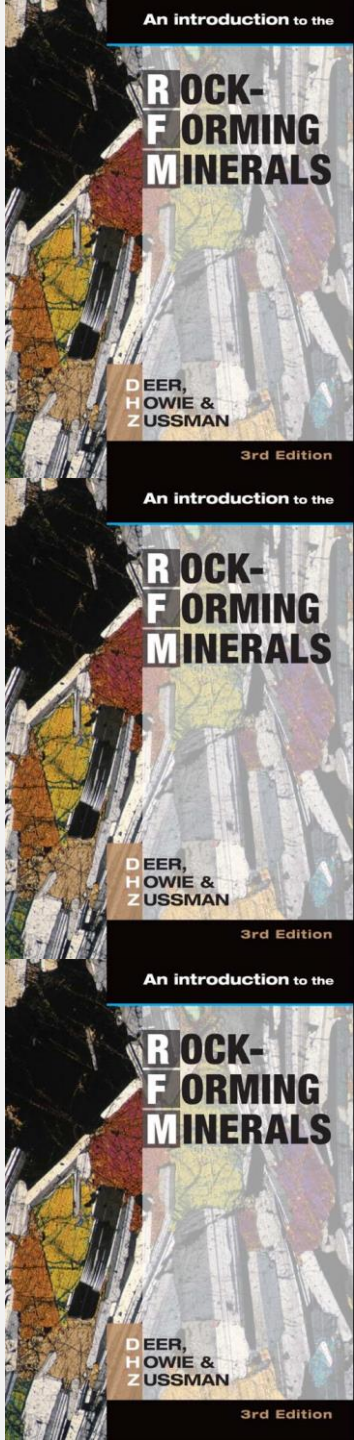




Found in:

Ultramafic igneous rocks
- Peridotite

Mafic igneous rocks
- basalt
- gabbro

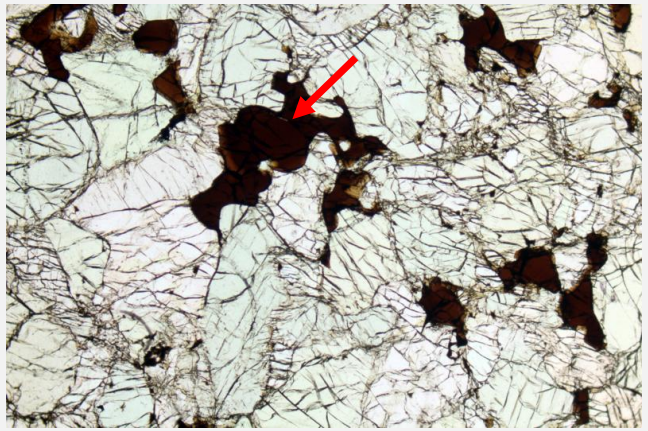


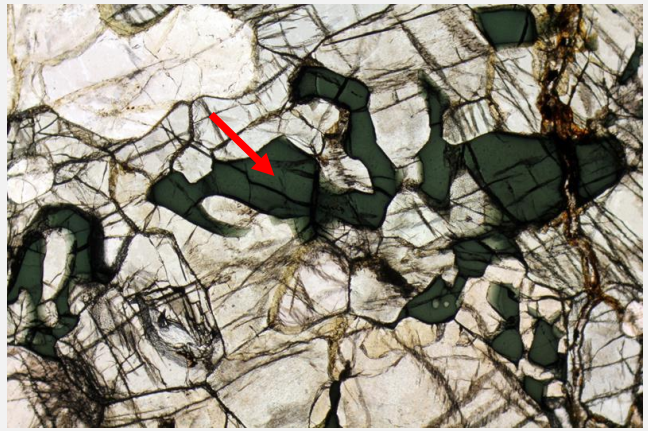
margaret.hartley@manchester.ac.uk

Spinel

Distinguish from garnet by its deep brown or green colour

Distinguish from other strongly coloured minerals by its isotropic extinction under crossed polars

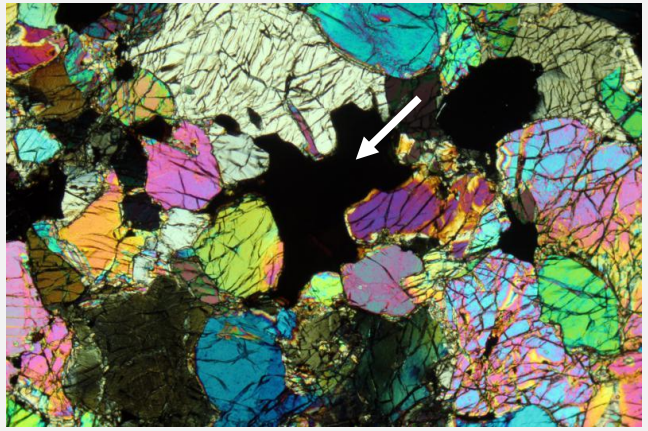


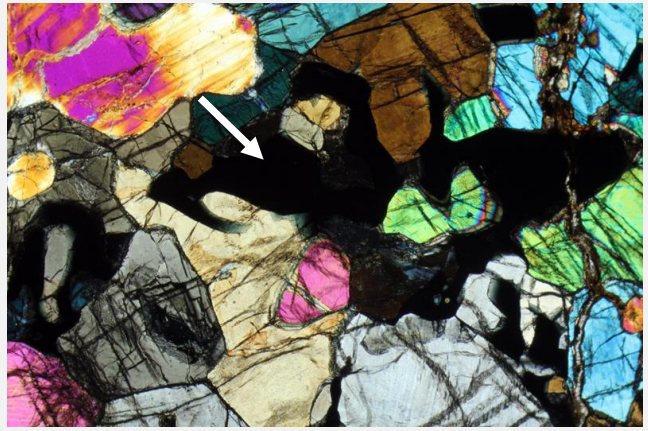


Key optical properties:

PPL

- Strongly coloured (often red, brown, green); may appear nearly opaque
- Not pleochroic
- High relief
- No cleavage





XP

- Isotropic – always in extinction
- No zoning

margaret.hartley@manchester.ac.uk

Magnetite

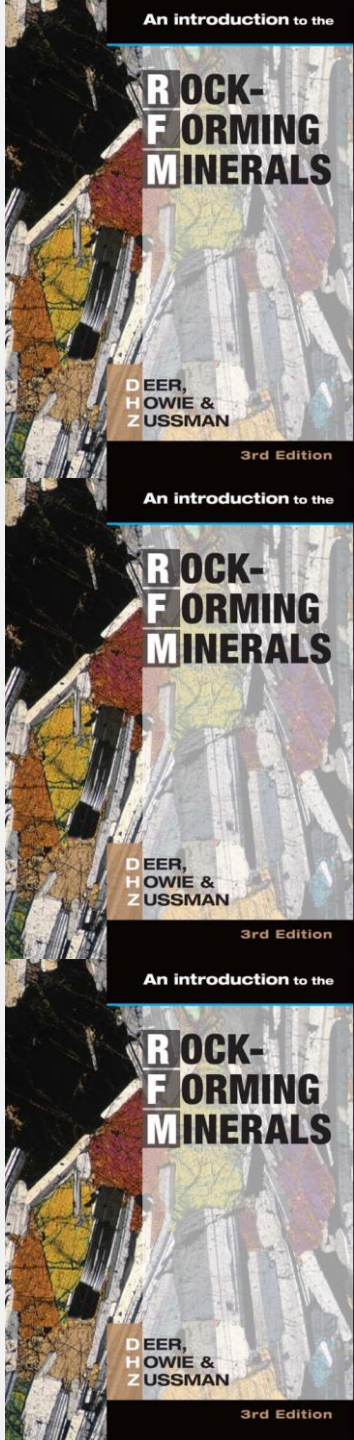
Fe_3O_4 oxide (non-silicate) cubic







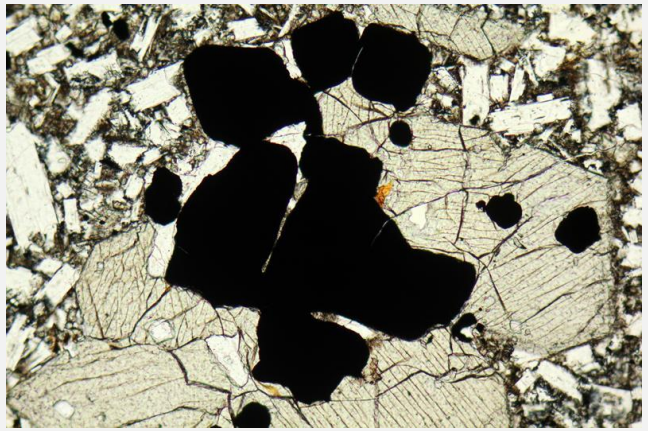
Found in a wide variety of mafic, intermediate and evolved igneous rocks

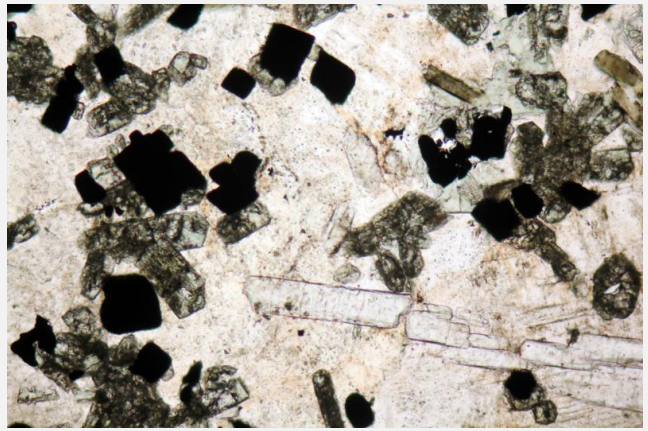


margaret.hartley@manchester.ac.uk

Magnetite

Distinguish from other minerals by its opaque appearance in PPL

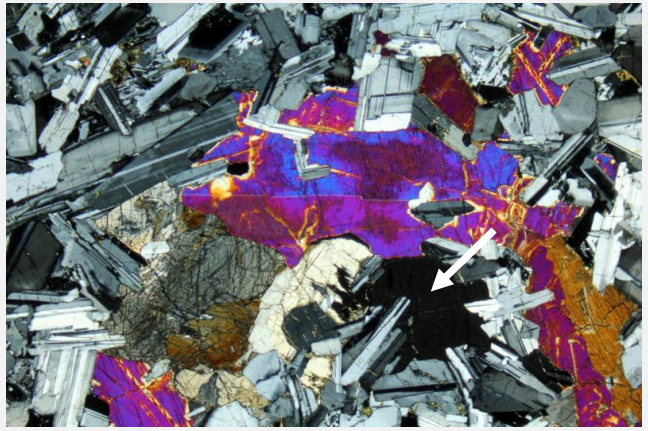


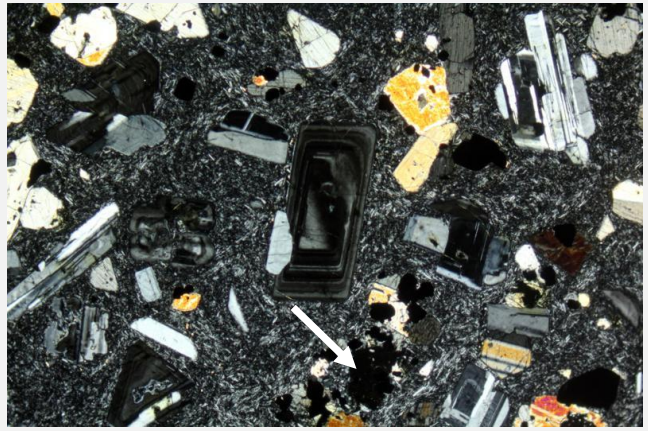


Key optical properties:

PPL

- Opaque
- Well-formed crystals may show equant cubic habit





XP

- Isotropic – always in extinction

margaret.hartley@manchester.ac.uk

Igneous processes: Melting

How do we know what the mantle is made of?

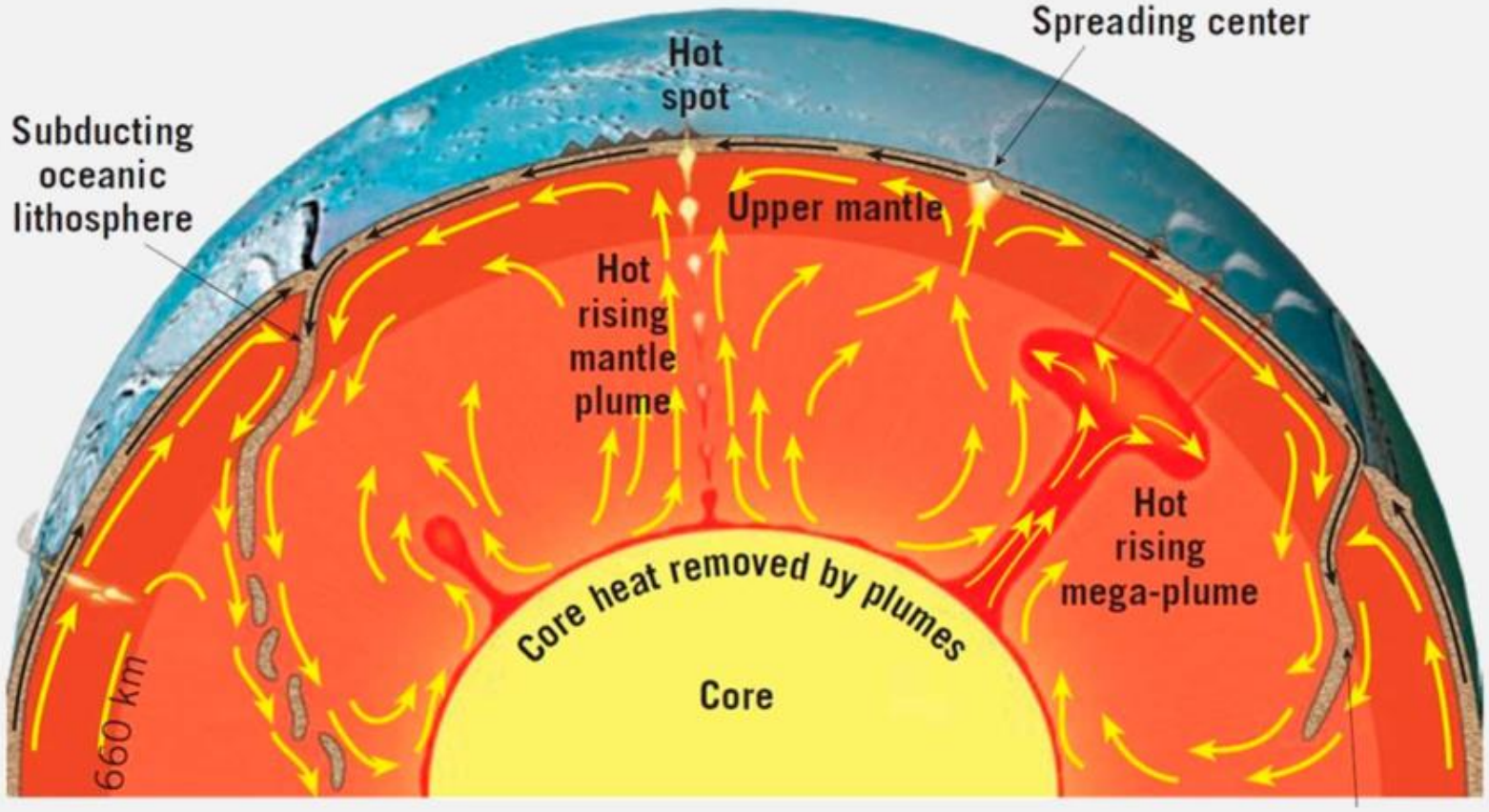
2.1 Indirect observations: constraints from geophysics

2.2 Direct observations from geological samples



margaret.hartley@manchester.ac.uk

How do we know what the mantle is made of?

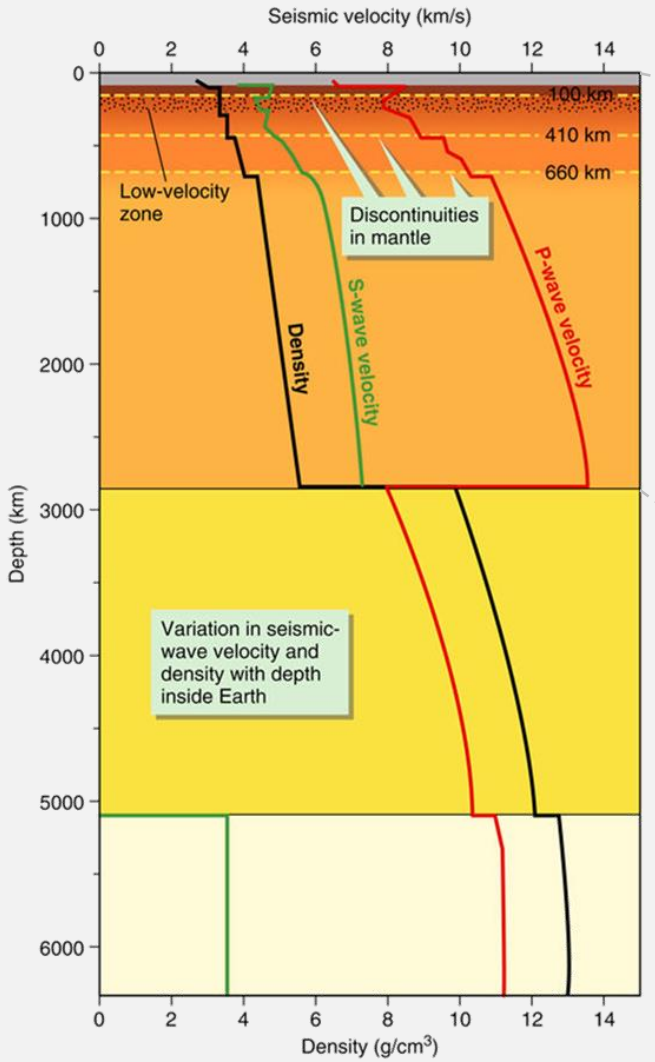


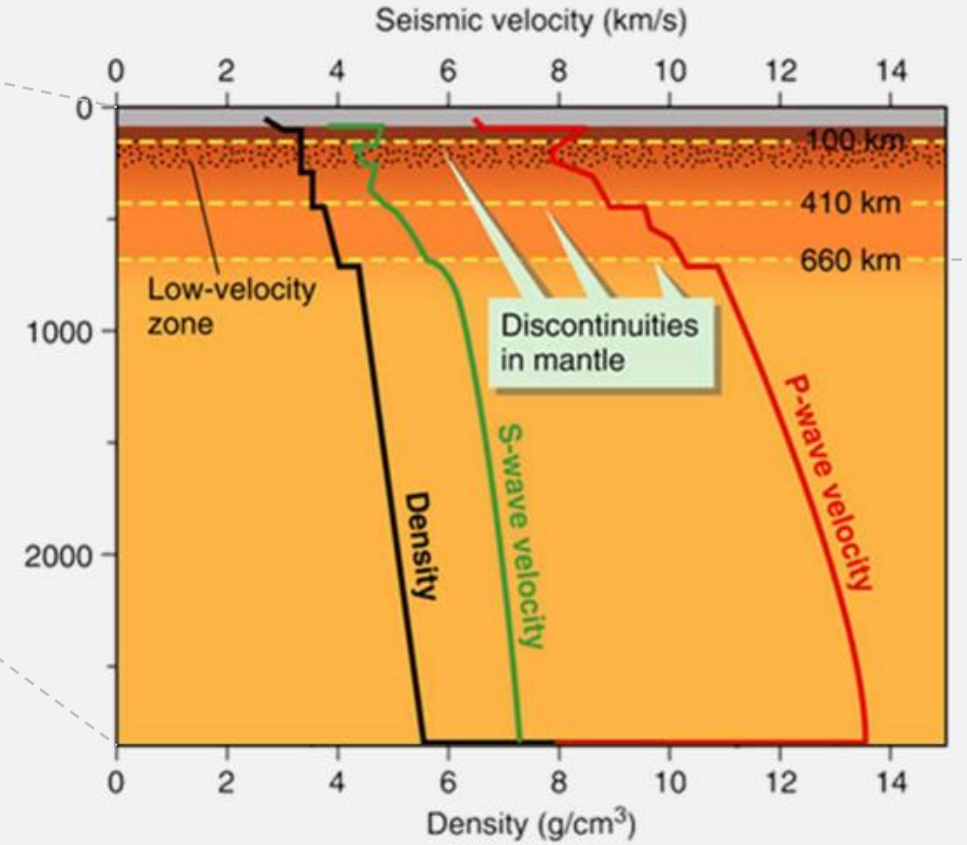
We know the structure, composition and mineralogy of the mantle through:

- Indirect geophysical observations
- Direct geological samples
- Petrological experiments

margaret.hartley@manchester.ac.uk

Geophysical observations: Mantle seismic velocity is a close match to olivine and pyroxene





upper mantle

lower mantle

margaret.hartley@manchester.ac.uk

Igneous processes: Melting

How do we know what the mantle is made of?

2.1 Indirect observations: constraints from geophysics

2.2 Direct observations from geological samples

- xenoliths

- orogenic peridotites

- ophiolite peridotites



margaret.hartley@manchester.ac.uk

Mantle xenoliths





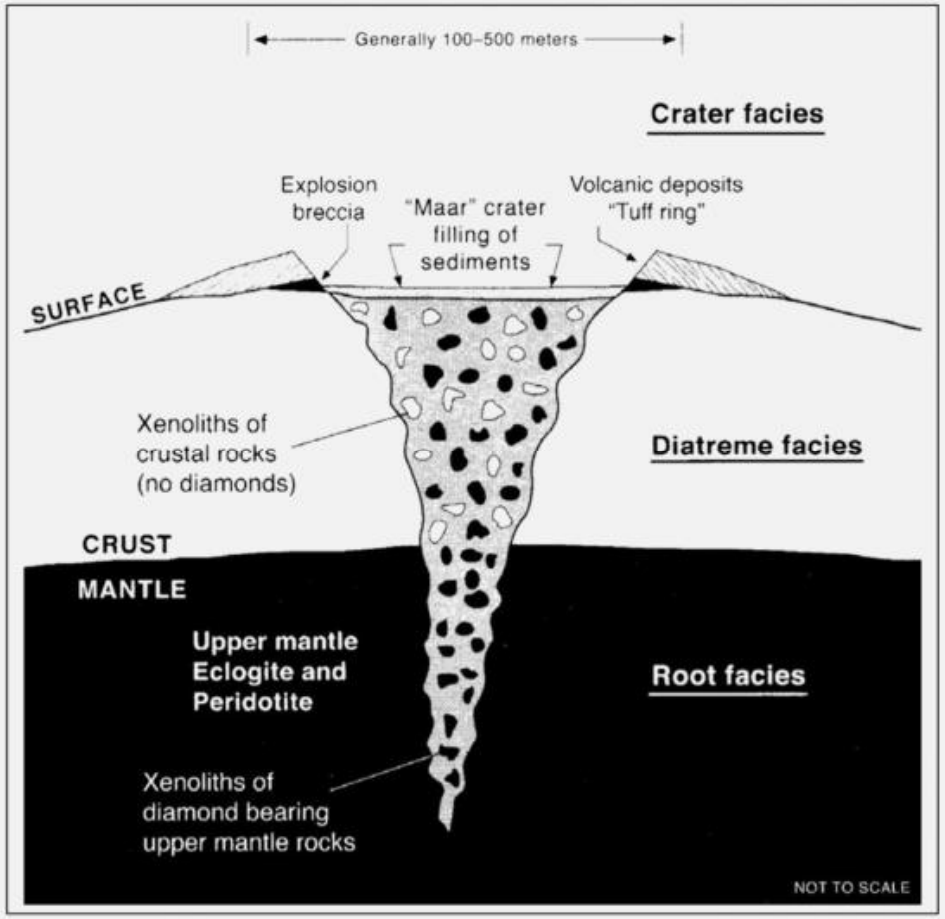
Mantle xenoliths are fragments of mantle rock that have been carried to the Earth's surface by an erupting magma

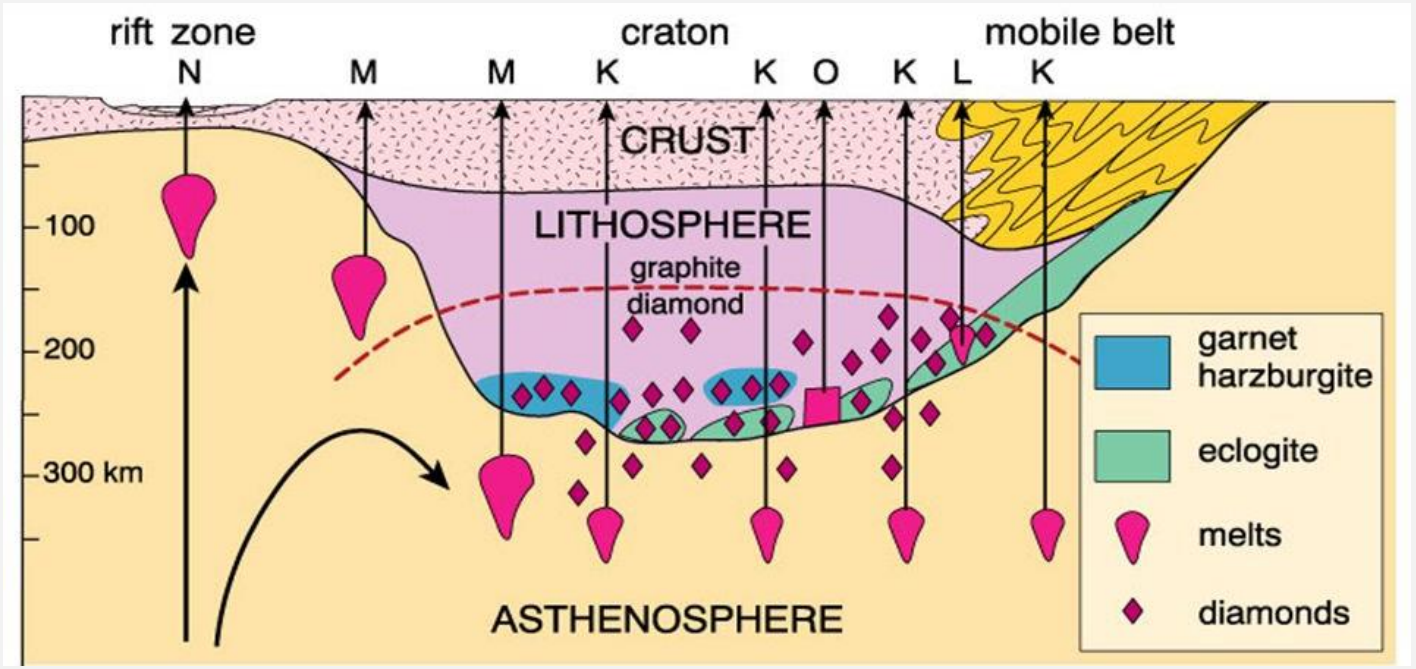


margaret.hartley@manchester.ac.uk

Mantle xenoliths are brought to the surface by kimberlite magmas

Kimberlite magmas form deep in the mantle. They ascend very rapidly at rates of 10-40 m/s.
They drill out xenoliths of dense mantle rock and transport them to the surface.





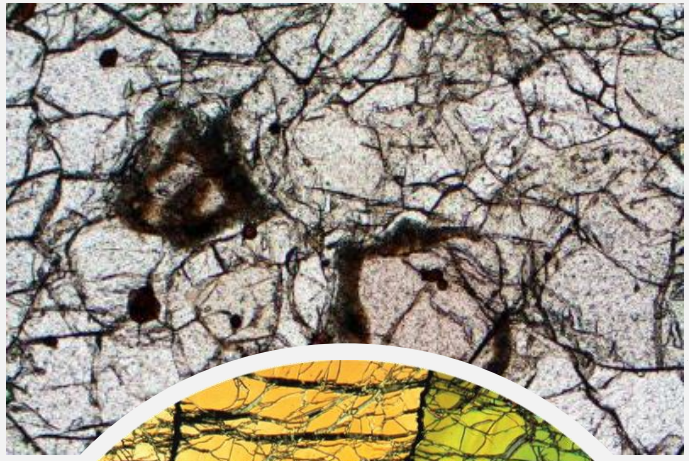
Mantle xenoliths from kimberlites are an economically important source of diamonds

margaret.hartley@manchester.ac.uk

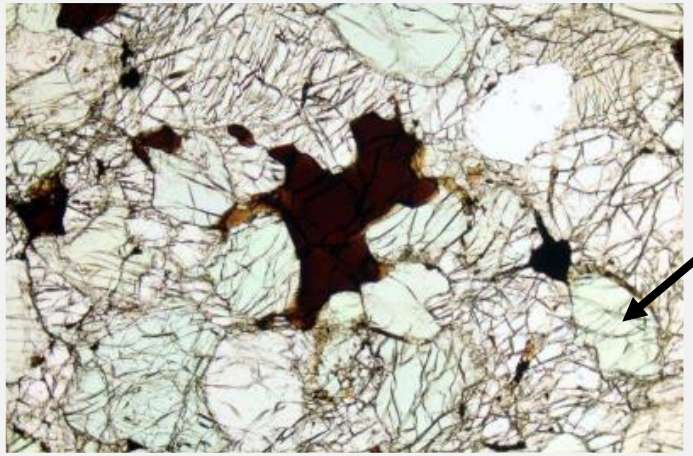
Garnet lherzolite

Spinel lherzolite

The aluminous phase reveals the pressure that the mantle xenolith came from

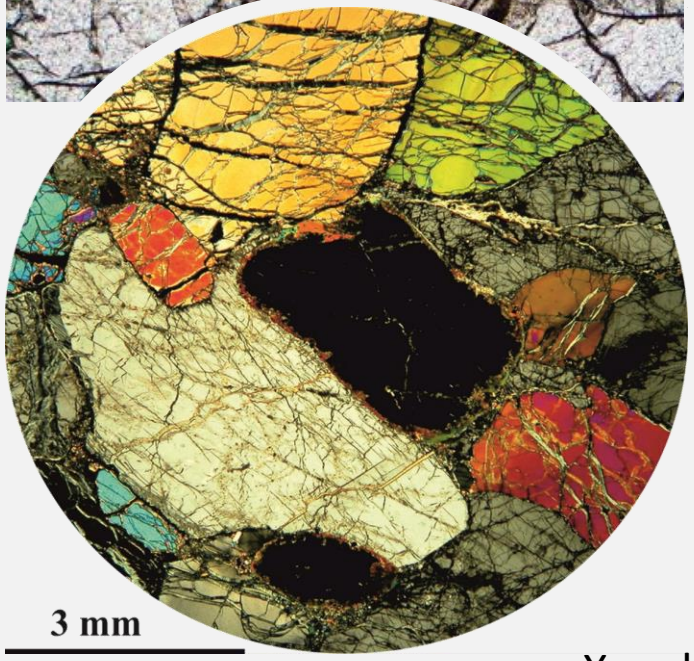


PPL

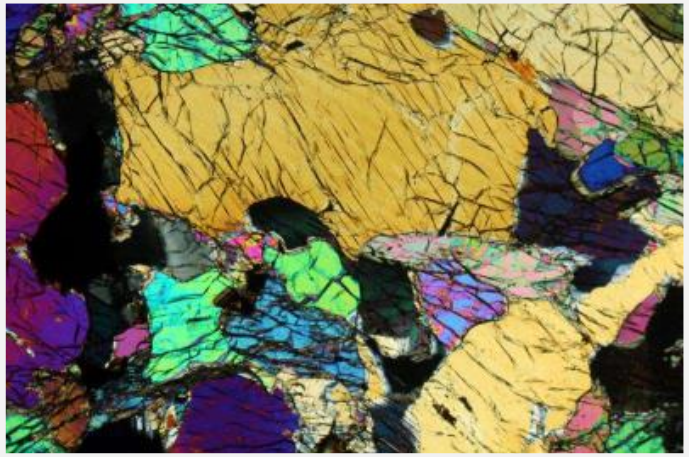


Cr-diopside





XP



Xenoliths often have a polygonal texture formed by solid-state diffusion and recrystallization processes. There is no textural evidence for the original order of crystallization.

margaret.hartley@manchester.ac.uk

Igneous processes: Melting

How do we know what the mantle is made of?

2.1 Indirect observations: constraints from geophysics

2.2 Direct observations from geological samples

- xenoliths

- orogenic peridotites

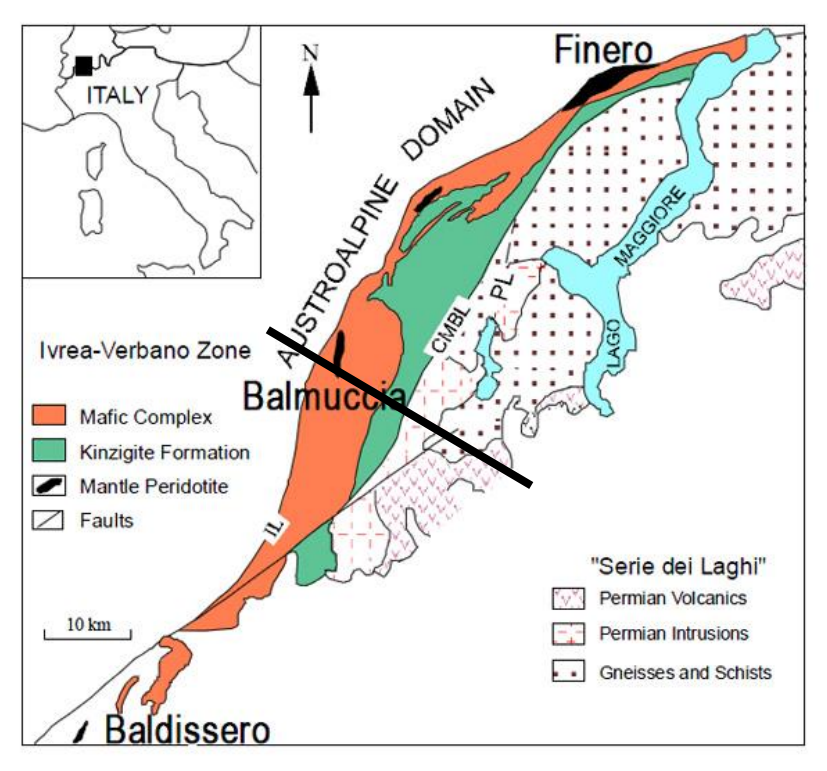
- ophiolite peridotites

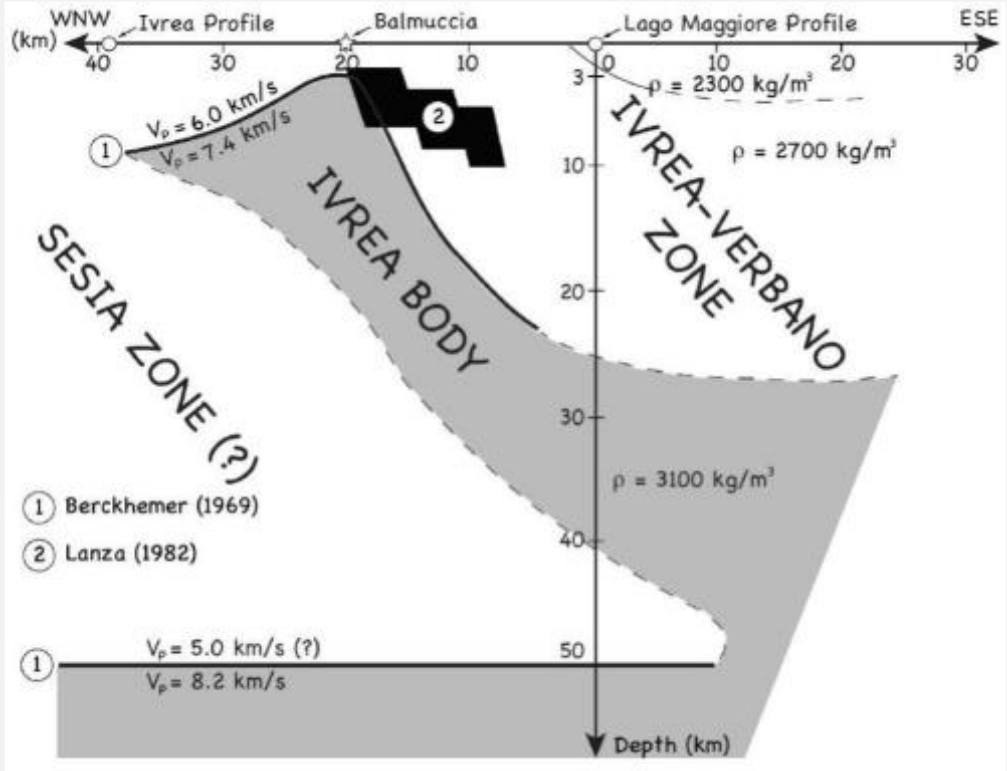


margaret.hartley@manchester.ac.uk

Orogenic peridotites are exhumed by major thrust faults

Example: the Ivrea-Verbano Zone in the Italian Alps





margaret.hartley@manchester.ac.uk

Orogenic peridotites: spinel lherzolite



Orogenic peridotite exposed in the Adula nappe, Swiss Alps



Alpine orogenic peridotites are spinel lherzolites, and often contain phlogopite.

margaret.hartley@manchester.ac.uk

Igneous processes: Melting

How do we know what the mantle is made of?

2.1 Indirect observations: constraints from geophysics

2.2 Direct observations from geological samples

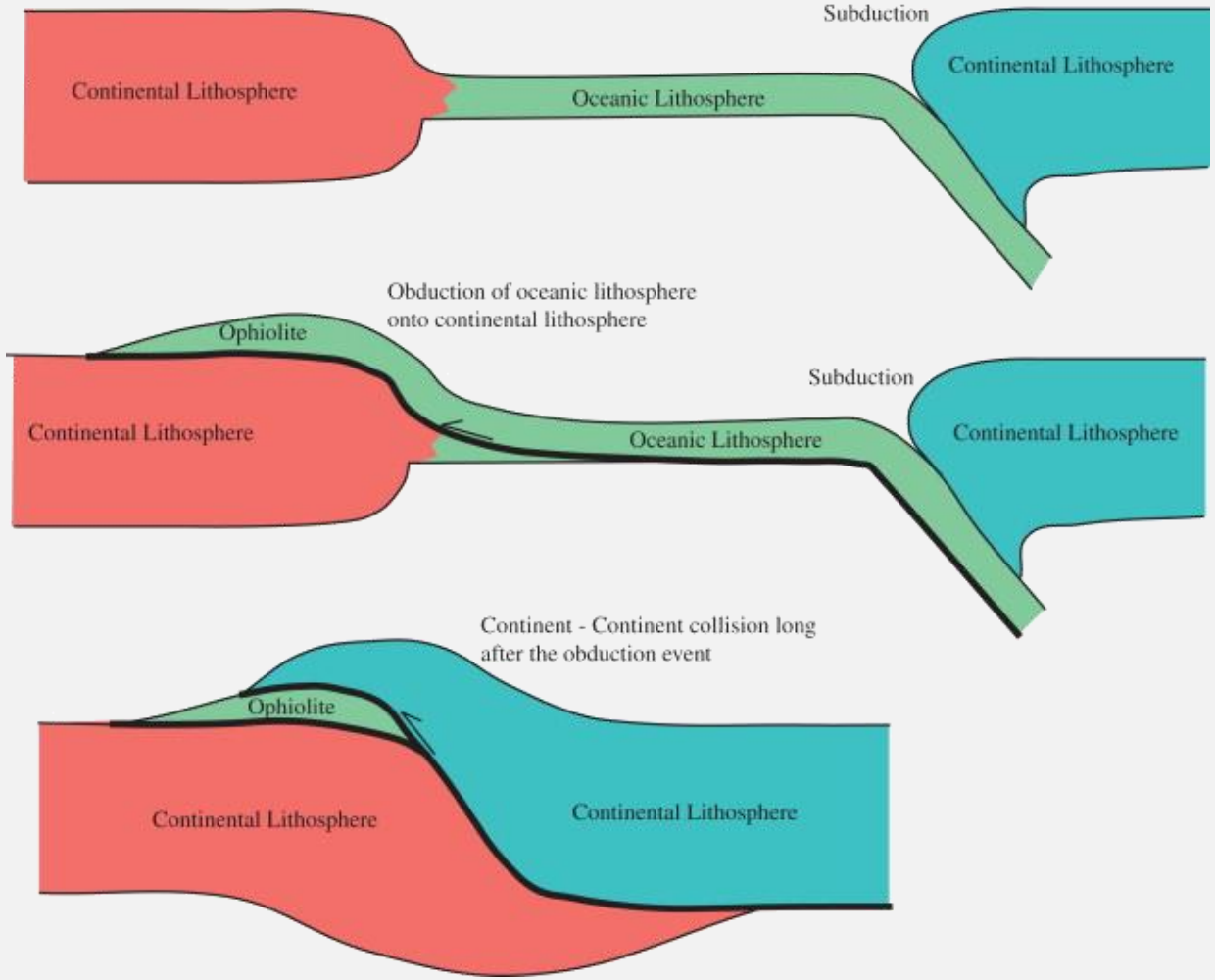
- xenoliths
- orogenic peridotites
- **ophiolite peridotites**

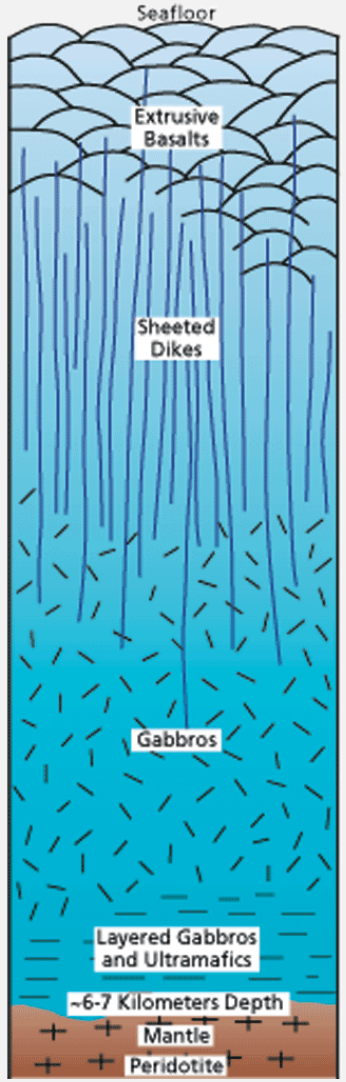


margaret.hartley@manchester.ac.uk

Ophiolites: oceanic lithosphere obducted onto continental lithosphere

More on the structure of the oceanic crust in Week 6





Mantle peridotites found at the very bottom of the ophiolite section, below the crustal rocks

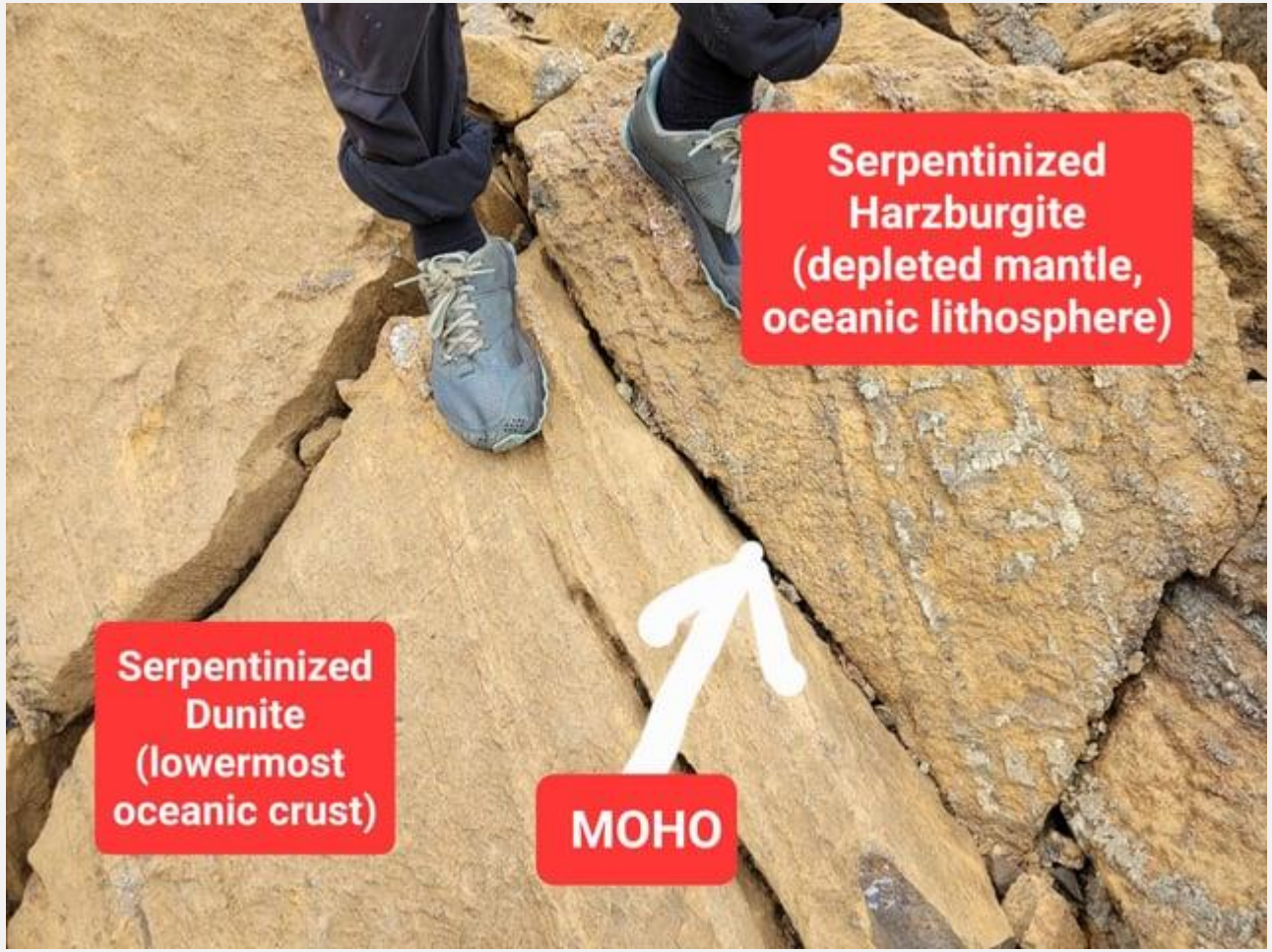
Seismic Moho

Petrologic Moho

margaret.hartley@manchester.ac.uk

Peridotites in ophiolites: harzburgite and dunite





Leka ophiolite complex, Norway

The **petrological Moho** separates mantle rocks (that have undergone melting) from crustal rocks (that have formed by crystallization of a magma)

Ophiolites can contain both mantle peridotites and crustal peridotites

Ophiolite peridotites are often highly serpentinized



margaret.hartley@manchester.ac.uk

Igneous processes: Melting

You should now be able to:

- Explain different forms of evidence for a peridotitic mantle
- Describe three different ways by which mantle rocks can be brought to the Earth's surface
- Compare and contrast the mineralogy of peridotites found in different tectonic contexts



margaret.hartley@manchester.ac.uk

Igneous processes: Melting

How does the mantle melt? What is produced, and what is left behind?

4.1 The peridotite ternary phase diagram

4.2 Equilibrium and fractional melting of the mantle

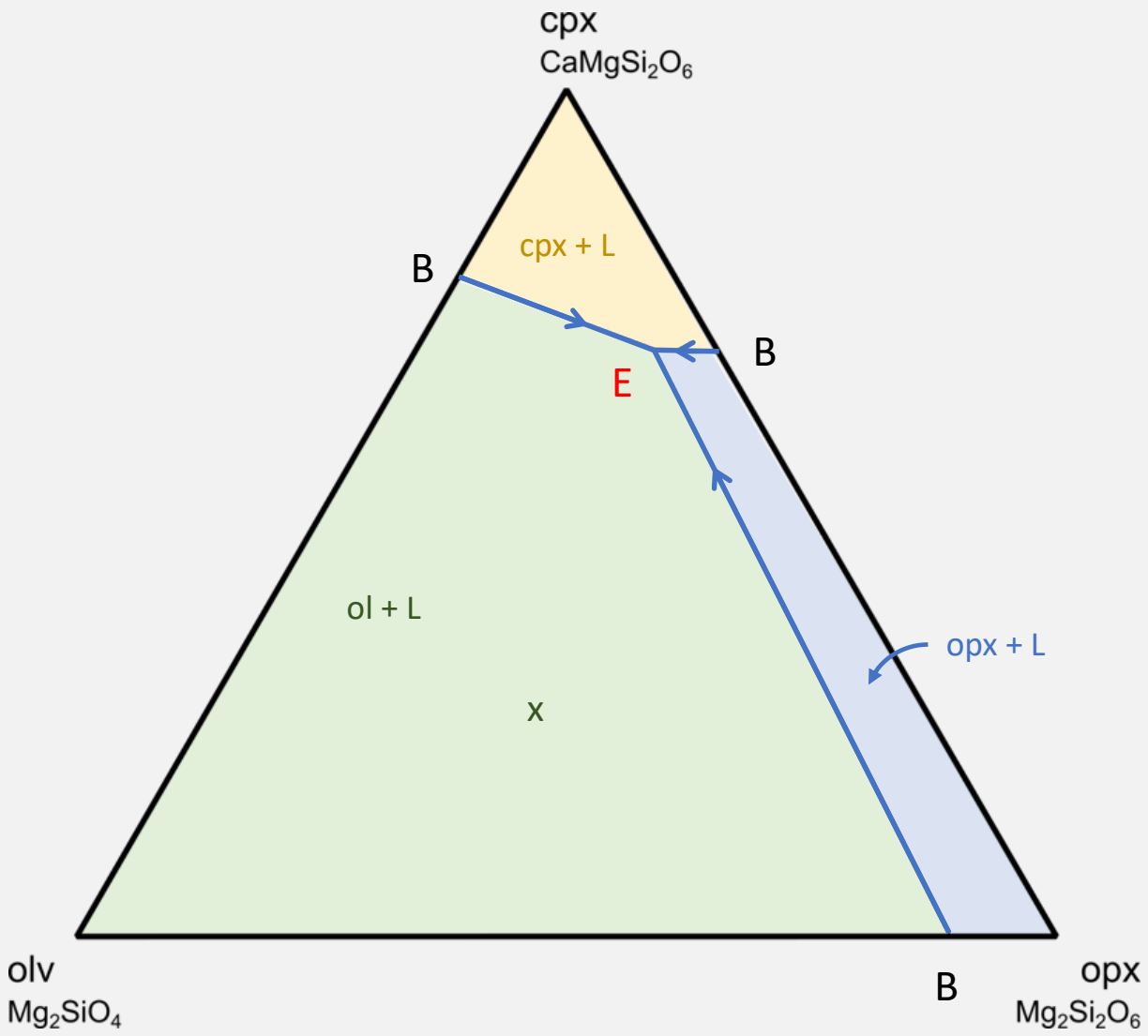
4.3 Fertile and depleted mantle

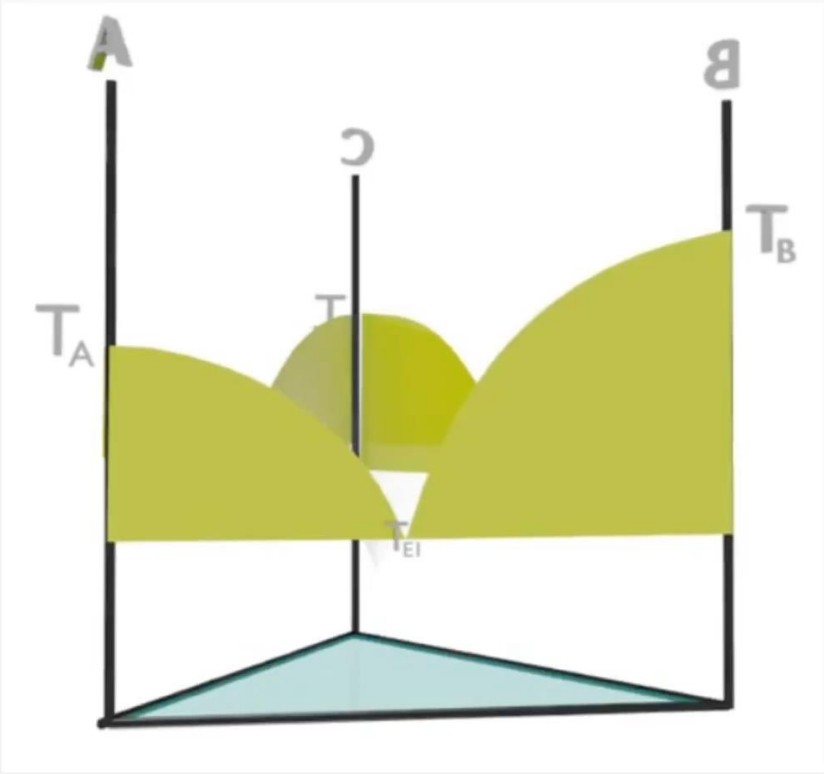
4.4 Compositions of primary mantle melts



margaret.hartley@manchester.ac.uk

Mantle melting on a ternary phase diagram





margaret.hartley@manchester.ac.uk

Igneous processes: Melting

How does the mantle melt? What is produced, and what is left behind?

4.1 The peridotite ternary phase diagram

4.2 Equilibrium and fractional melting of the mantle

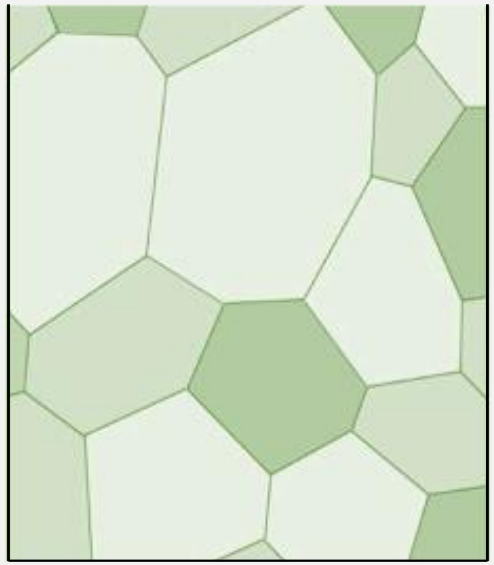
4.3 Fertile and depleted mantle

4.4 Compositions of primary mantle melts

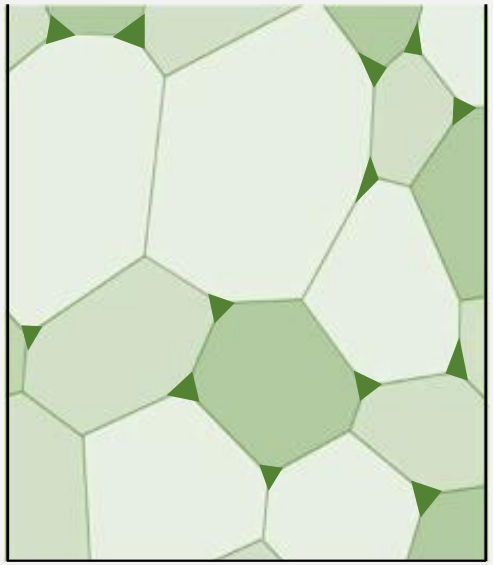


margaret.hartley@manchester.ac.uk

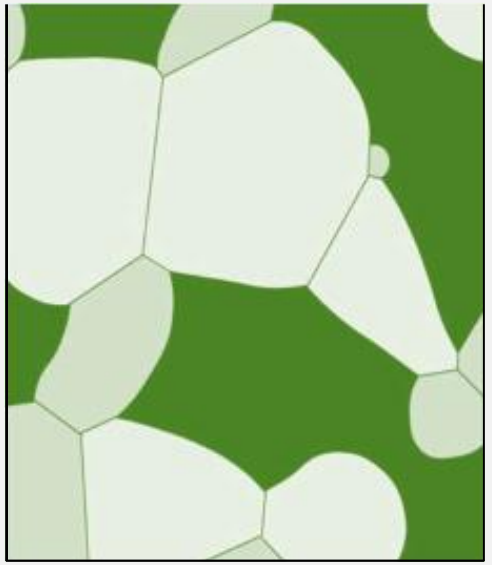
Equilibrium melting



0% melting
0% melt



5% melting
5% melt

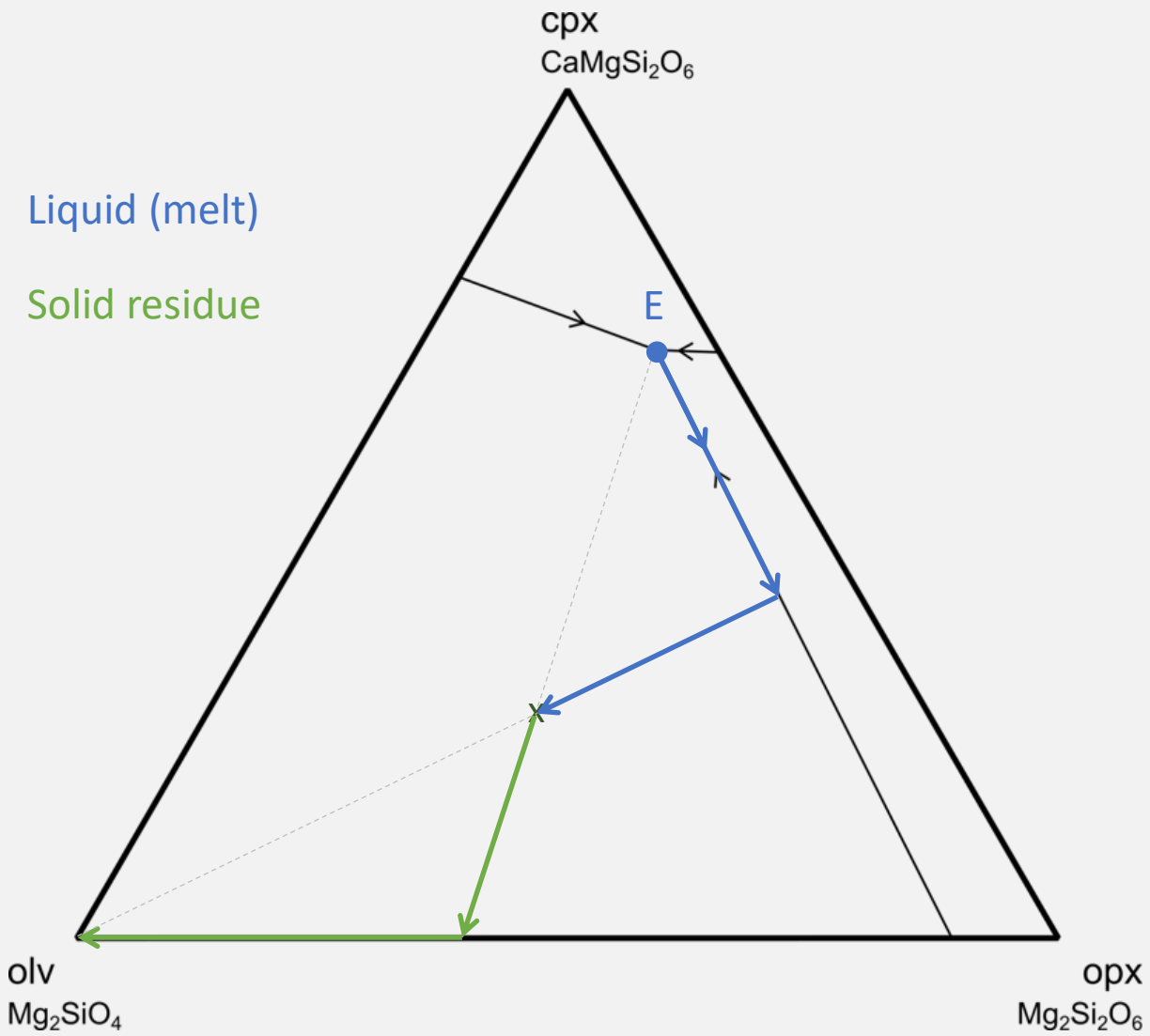


10% melting
10% melt

melting

margaret.hartley@manchester.ac.uk

Equilibrium melting



First melt at ternary eutectic:

olv + cpx + opx → L

Cpx runs out. New melting reaction:

olv + opx → L

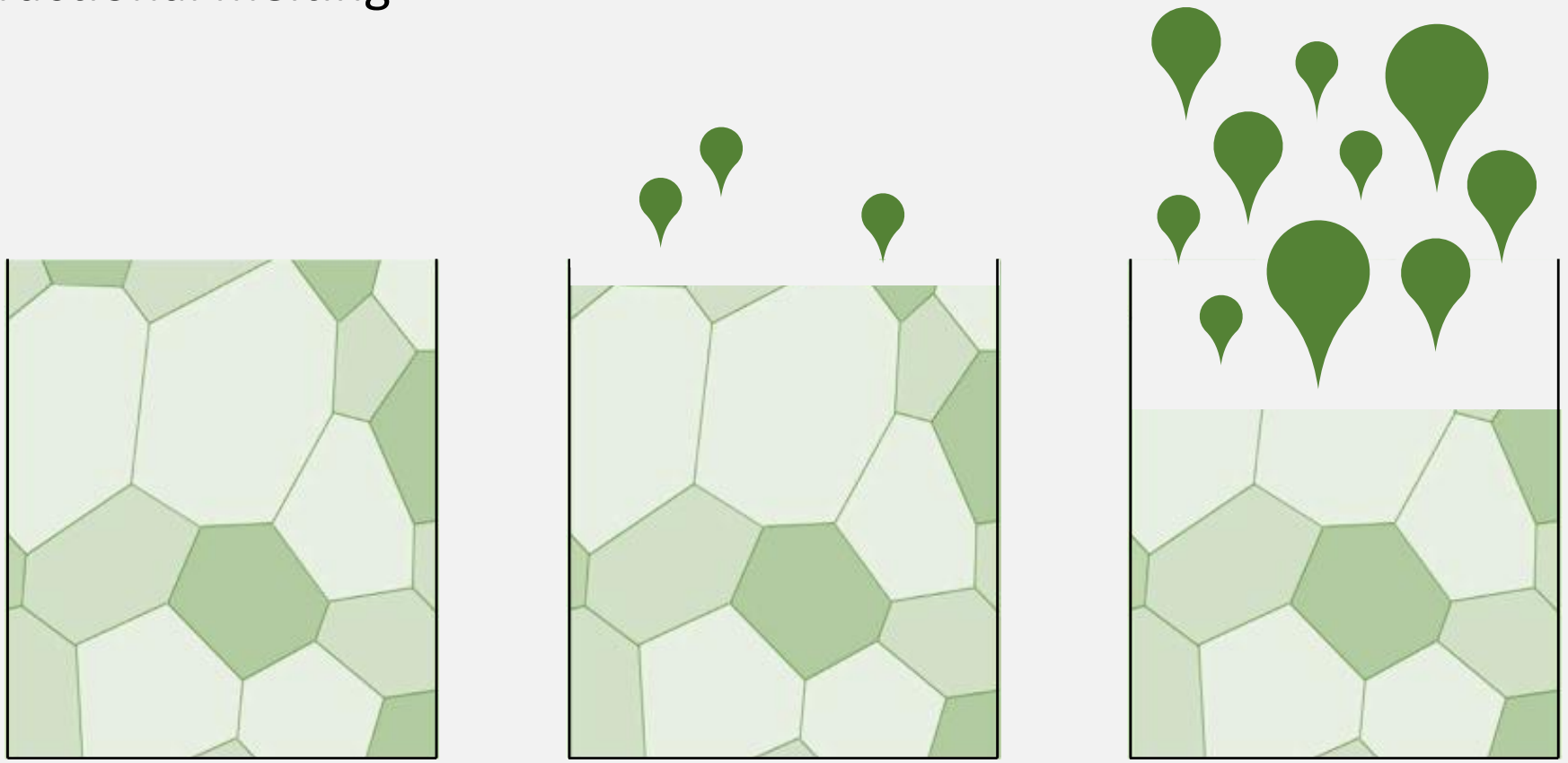
Opx runs out. New melting reaction:

olv → L

Final melt has same composition as initial solid

margaret.hartley@manchester.ac.uk

Fractional melting



0% melting
0% melt

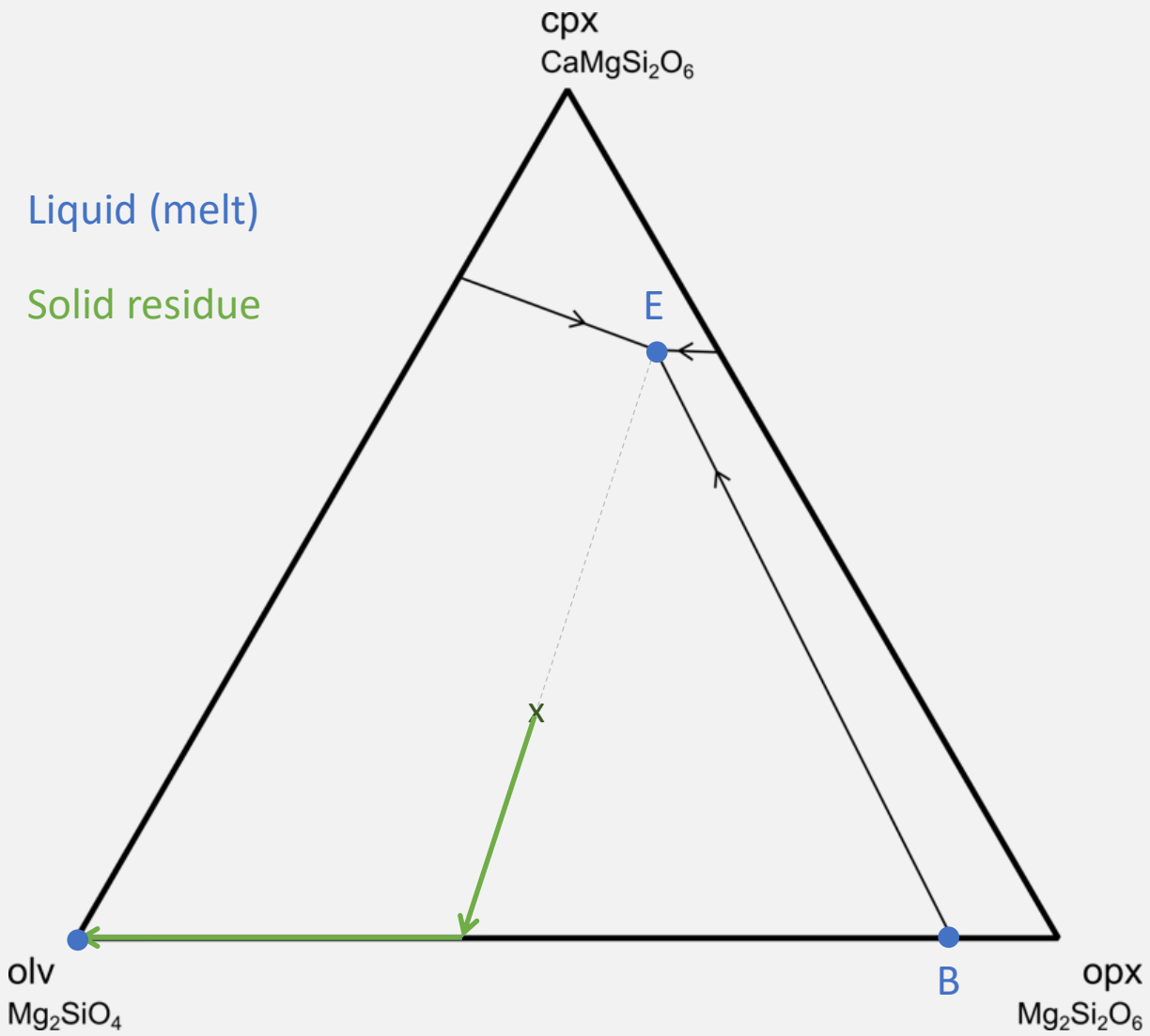
5% melting
0% melt

10% melting
0% melt

melting

margaret.hartley@manchester.ac.uk

Fractional melting



First melt at ternary eutectic:

olv + cpx + opx → L

Cpx runs out. Melting stops.

Second melt at binary eutectic:

olv + opx → L

Opx runs out. Melting stops.

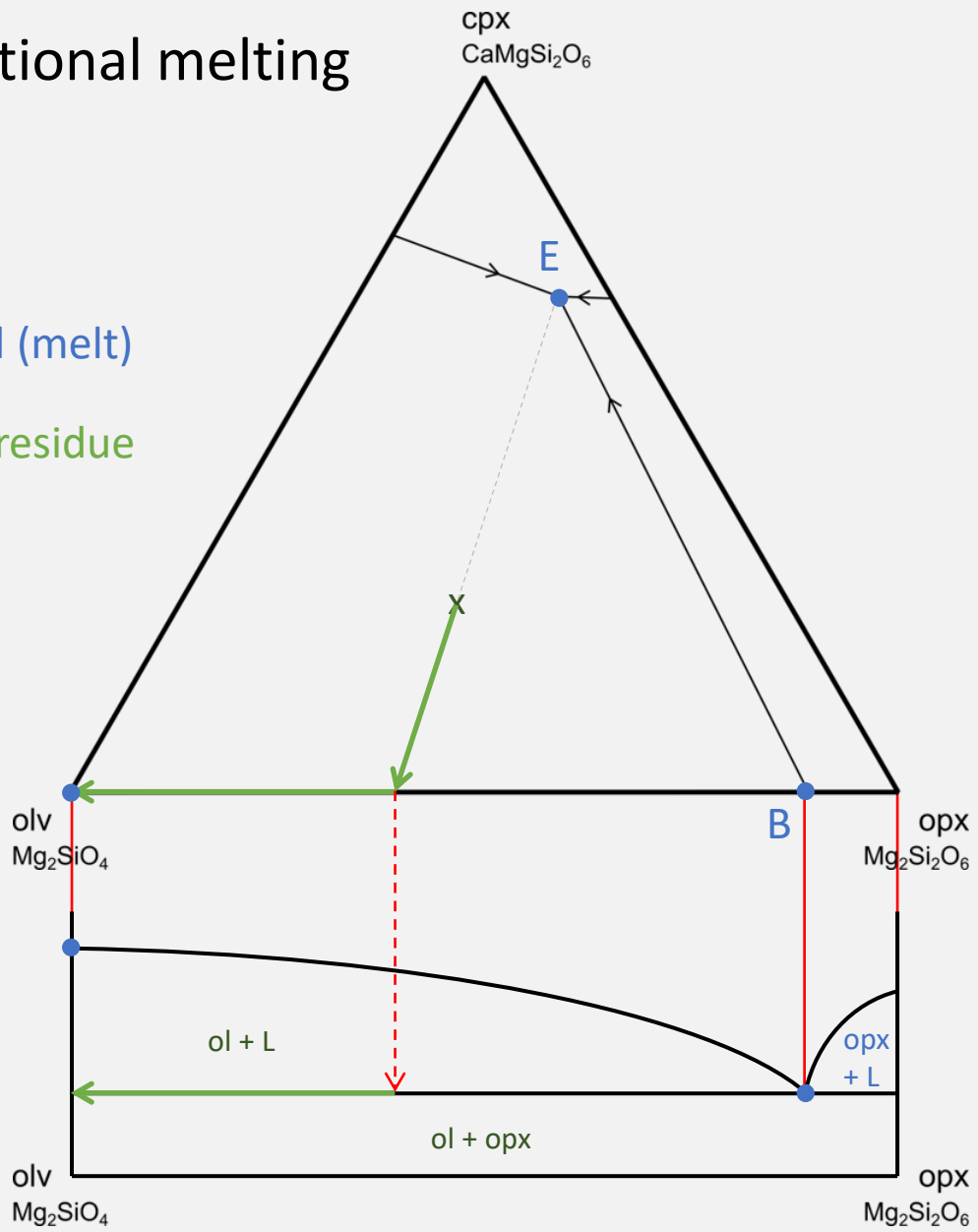
Third melting reaction:

olv → L

Final melt has the composition of pure olivine

margaret.hartley@manchester.ac.uk

Fractional melting



See also EART 26201
Principles of Geochemistry

First melt at ternary eutectic:

olv + cpx + opx → L

Cpx runs out. Melting stops.

Second melt at binary eutectic:

olv + opx → L

Opx runs out. Melting stops.

Third melting reaction:

olv → L

Final melt has the composition of pure olivine

margaret.hartley@manchester.ac.uk

Igneous processes: Melting

How does the mantle melt? What is produced, and what is left behind?

4.1 The peridotite ternary phase diagram

4.2 Equilibrium and fractional melting of the mantle

4.3 Fertile and depleted mantle

4.4 Compositions of primary mantle melts



margaret.hartley@manchester.ac.uk

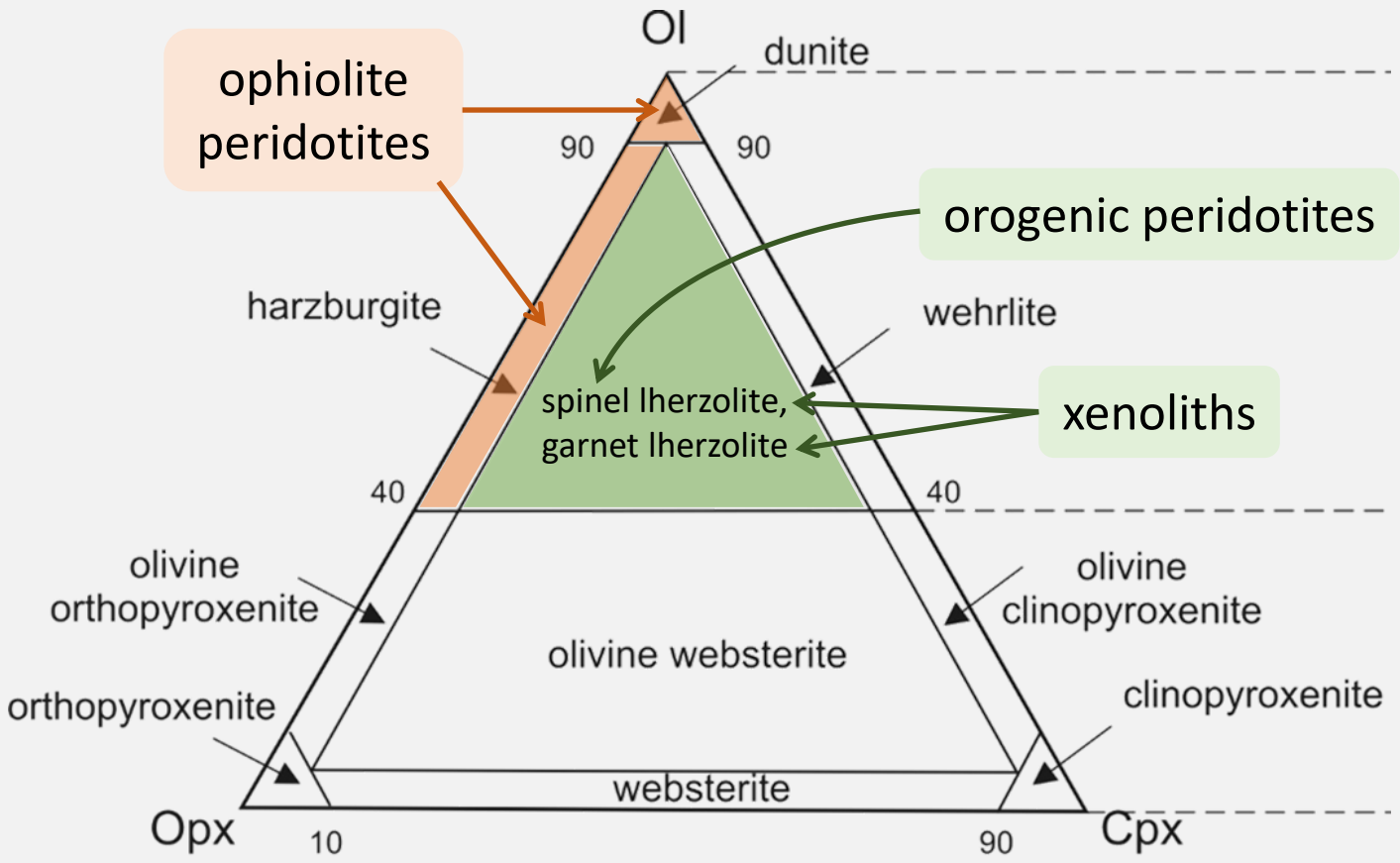
Fertile vs depleted mantle

fertile mantle

- Has undergone no (or very little) previous melting
- Ol + cpx + opx + Al-bearing phase
- Incompatible elements are still present

depleted mantle

- Has previously undergone melting
- Ol + opx; contains no (or very little) cpx or Al-bearing phase
- Incompatible elements have been stripped out by past melting



margaret.hartley@manchester.ac.uk

Igneous processes: Melting

How does the mantle melt? What is produced, and what is left behind?

4.1 The peridotite ternary phase diagram

4.2 Equilibrium and fractional melting

4.3 Fertile and depleted mantle

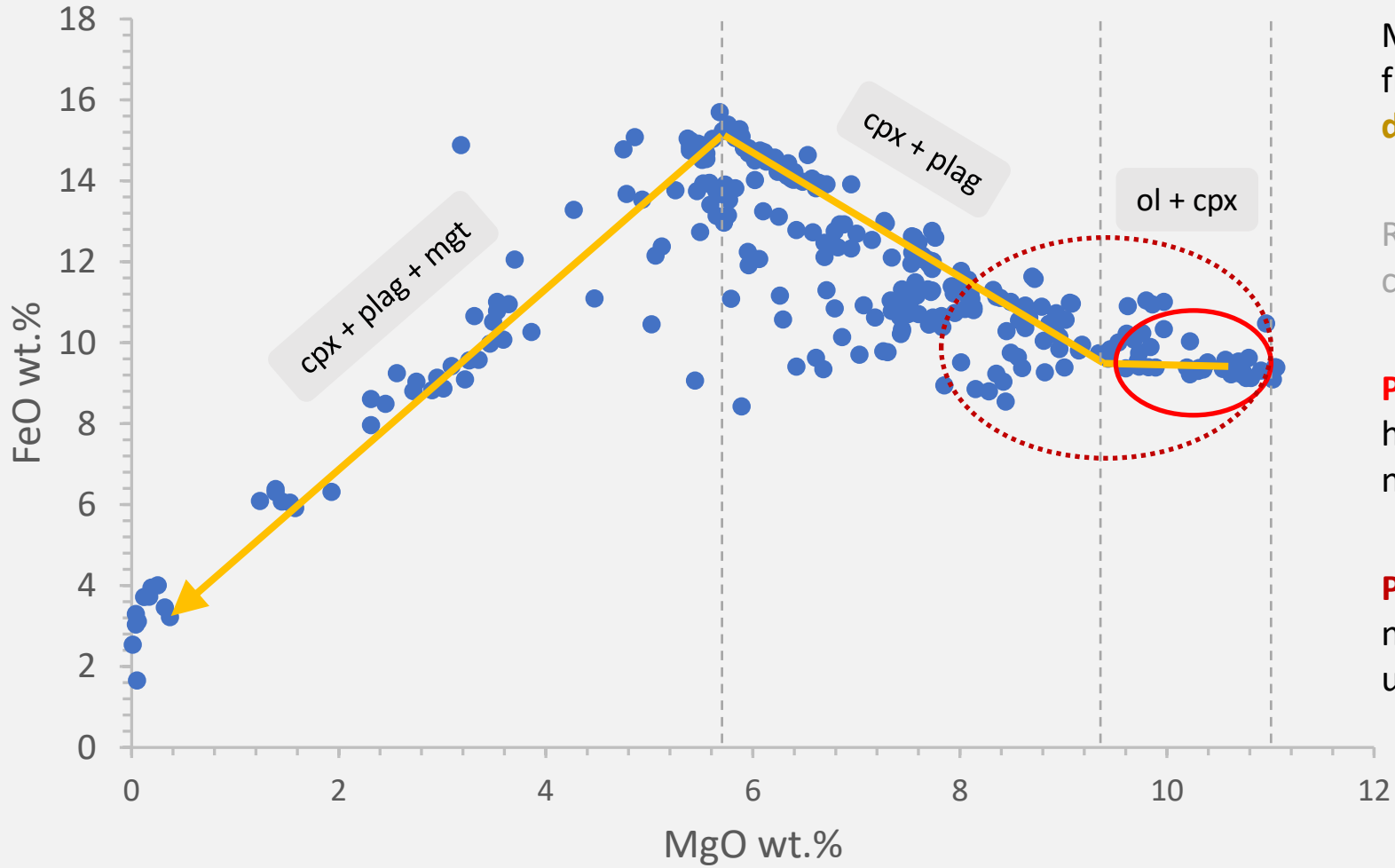
4.4 Compositions of primary mantle melts



margaret.hartley@manchester.ac.uk

What are primary melts?

This is the liquid line of descent from Practical 4



Magma compositions related to one another by fractional crystallization through a **liquid line of descent (LLD)**

Reminder: changes in gradient on the LLD indicate changes to the crystallizing mineral assemblage

Primary melts are derived from the mantle. They have very high MgO contents and have undergone no (or very little) fractional crystallization.

Primitive melts have high MgO contents, but are not necessarily primary – they may have undergone some fractional crystallization

margaret.hartley@manchester.ac.uk

Compositions of primary melts vs mantle peridotites

wt.%	lherzolite peridotite	Primary melt 1	Primary melt 2	Primary melt 3
SiO_2	45.62	49.16	51.02	46.59
TiO_2	0.20	2.29	2.03	2.26
Al_2O_3	2.55	13.33	13.49	15.19
FeO	7.59	10.89	11.02	12.55
MnO	0.13	0.16	0.17	0.18
MgO	41.41	10.41	8.42	8.74
CaO	1.90	10.93	10.30	10.02
Na_2O	0.19	2.15	2.10	3.01
K_2O	0.12	0.51	0.40	0.96
P_2O_5	0.03	0.16	0.26	0.29

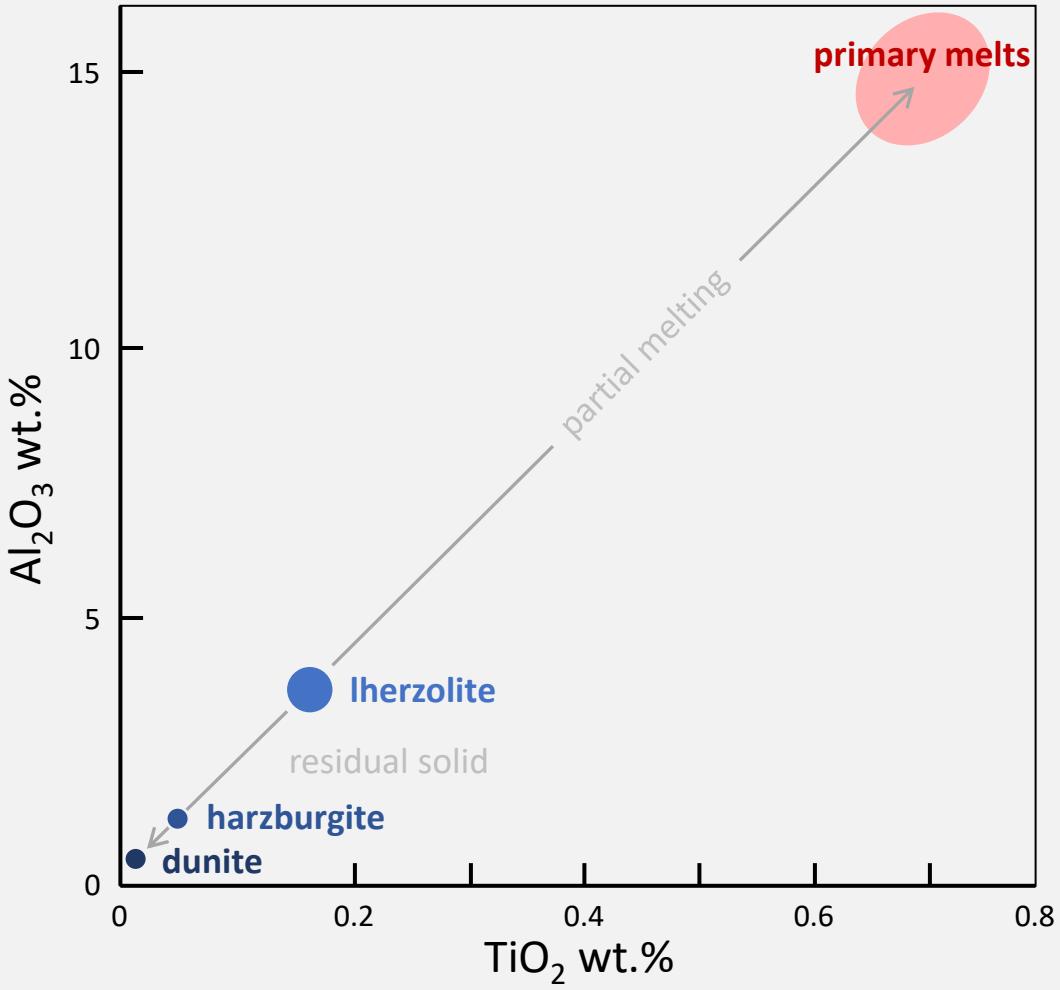
As melting proceeds:

- Incompatible elements are preferentially partitioned into the melt
- Compatible elements are preferentially retained in the residual solid

The primary melts are all basalts

margaret.hartley@manchester.ac.uk

Compositions of mantle rocks vs primary melts



Primary melts are enriched in incompatible elements compared to the starting mantle composition

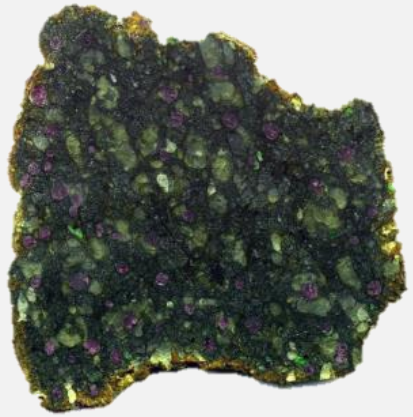
When mantle lherzolite undergoes partial melting, incompatible elements preferentially enter the melt phase. Al and Ti are both incompatible elements

The residual solid left over after melting is depleted in incompatible elements compared to the starting mantle composition

margaret.hartley@manchester.ac.uk

The exact primary melt composition depends on the melting conditions: P, T and F

P = pressure
T = temperature
F = melt fraction



Garnet peridotite melting at 2 GPa

1450 °C: 1% melting — alkali basalt

1500 °C: 16% melting — tholeiitic basalt



Spinel peridotite melting at 1 GPa

1350 °C: 15% melting — alkali basalt

1450 °C: 20-30% melting — tholeiitic basalt

Tholeiitic basalts are formed by shallower melting and greater % partial melting (typically 15-30%).

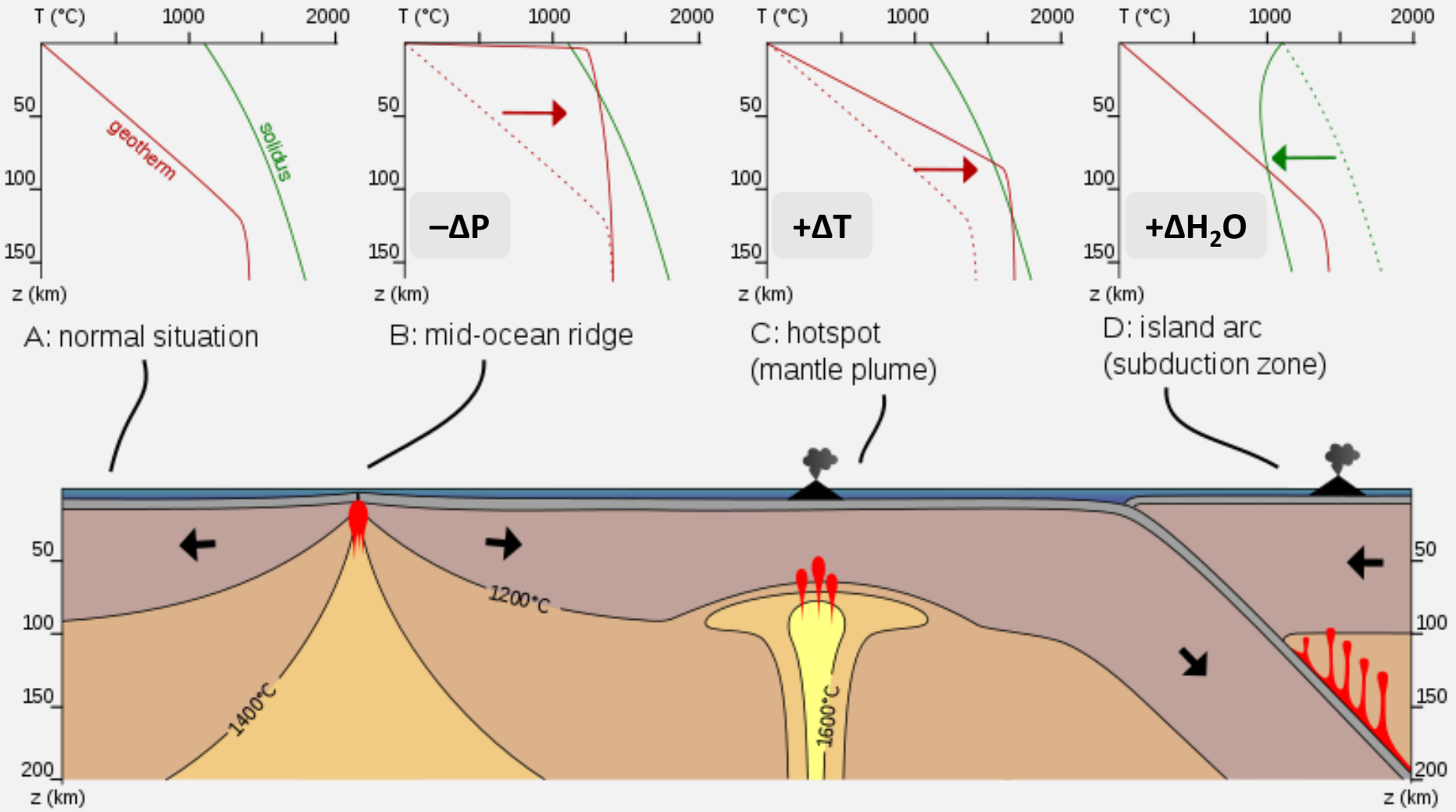
Alkali basalts are formed by deeper melting and lower % partial melting (typically <10%).

We will return to these ideas in weeks 6-8

margaret.hartley@manchester.ac.uk

Where and how are different primary melt compositions produced?

To be continued in weeks 6, 7 and 8



margaret.hartley@manchester.ac.uk

Igneous processes: Melting

You should now be able to:

- Explain the difference between equilibrium and fractional melting
- Illustrate the process of mantle melting using a ternary phase diagram
- Explain what is meant by fertile and depleted peridotite
- Explain how different mantle rock types are related to one another through partial melt extraction
- Outline the compositional differences between mantle rocks and primary melts



margaret.hartley@manchester.ac.uk